

## Solution Design Document

SIMF, Saudi International Maritime Forum

Low-Level Design, SIMF-LLD-004, Version 2.0

### DOCUMENT INFORMATION

| Sr. No. | Date       | Ver. | Name                                 | Action | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------|------------|------|--------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | 2026-07-21 | 1.0  | SIMF Engineering & Architecture Team | Create | Initial issue of the Solution Design Document, authored to the DoD Low-Level Design template from the SIMF design baseline and the source tree.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2       | 2026-08-10 | 1.1  | SIMF Engineering & Architecture Team | Update | Added the component, data flow and deployment figures. Corrected the real-time transport statements after the reserved server-push component was removed from the solution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3       | 2026-08-11 | 1.1  | SIMF Engineering & Architecture Team | Update | Third-party services shown as data flow rather than as architecture, with the data each receives itemised.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4       | 2026-08-12 | 1.2  | SIMF Engineering & Architecture Team | Update | Replaced the deployment figure with the target three-tier separation diagram. Answered the owner clarifications: audience, data classification, administrators, external integrations, website scope, video library and AI capabilities.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 5       | 2026-08-20 | 1.3  | SIMF Engineering & Architecture Team | Update | Replaced the deployment figure with the security areas and internet egress sheet. The zones are grouped into HSA and SSA, a load balancer sits in front of the API nodes, and the API calls two internet services outbound through both firewalls.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6       | 2026-08-30 | 1.4  | SIMF Engineering & Architecture Team | Update | Phase one. The API server moves into HSA and loses its internet path; the AI moves from a cloud provider to an on-site LLM server in HSA; mail moves to an on-site relay in HSA; the file store moves from a directory on a file share to MinIO object storage in HSA; and the one remaining internet call, the YouTube caption fetch, is made from the Control Panel in SSA.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7       | 2026-08-30 | 1.5  | SIMF Engineering & Architecture Team | Update | Aligned the text with the deployment diagram. Gemini is replaced by the SITE-hosted GPT OSS 120B model on an on-site LLM server in HSA; the file store is MinIO over the S3 API; the separate background worker is removed, the API running its jobs in-process; and sign language and live translation are withdrawn. Made the document self-contained: every reference to an artefact outside the delivered set removed, section 1.4 listing the three delivered documents, and an annex mapping all 132 business requirement identifiers to the sections that design them. Corrected the Website screens table, which listed nine sign-in and account routes the site does not carry and omitted thirteen it does. Secret custody now reads the same here as in the High Level Design. |
| 8       | 2026-09-04 | 2.0  | SIMF Engineering & Architecture Team | Update | Rebuilt as a design document. Requirement traceability moves from an annex to numbered section 8, with a verification method against every identifier. An entity relationship diagram is added to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Sr. No. | Date | Ver. | Name | Action | Description                                                                             |
|---------|------|------|------|--------|-----------------------------------------------------------------------------------------|
|         |      |      |      |        | section 6.2. References to artefacts outside the three delivered documents are removed. |

## Overview

This Solution Design Document is the detailed technical design for SIMF, the Saudi International Maritime Forum. It follows the DoD Solution Design Document structure, describes the delivered system, and is the baseline for development, review and quality assurance. It carries the database schema, interface specification and unit-level logic.

The document is self-contained: where a section catalogues a set, the whole set is given. Section 3 holds the use cases, section 4 the client screens and routes, section 5 the modules, and section 6 the entities and tables.

Section 8 traces all 132 requirement identifiers in the Business Requirements Document to the implementing module and to a verification method. Four are recorded there as withdrawn.

The only cited documents are the two delivered alongside it: the Business Requirements Document, SIMF-BRD-001, and the High Level Design, SIMF-HLD-005, both at version 2.0. Section 1.4 lists all three.

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## 1. Introduction

### 1.1 Purpose

This Solution Design Document, SIMF-LLD-004, defines the technical solution design for SIMF, the Saudi International Maritime Forum platform. It sets out how the system is structured, the classes and interfaces that implement each concern, the request pipeline, the database schema, the client applications, the shared libraries, and the runtime flows that connect them. It follows the Ministry of Defense Solution Design Document template: section 7 is the Solution Description, and section 8 carries requirement traceability.

The intended audience is the engineering team that implements and maintains SIMF, meaning its architects and developers, the QA function that verifies it, and the Ministry of Defense reviewer who assesses the design. This document is the reference baseline against which development is carried out and against which the delivered system is reviewed.

This document describes the technical design. It does not restate the business requirements, which are carried by the Business Requirements Document delivered alongside it. Section 8 traces every requirement identifier to the section here that designs it and states the verification method for each one. Where a low-level detail is not settled, this document marks it as open rather than inventing a value.

### 1.2 Scope

This design covers the whole SIMF platform. The component set is:

- Backend API: SIMF.Api, a .NET 10 and FastEndpoints HTTP host with route prefix api/v1, layered as Domain, Application, Infrastructure, Api.

- Control Panel: a Blazor Server administrative application over the /admin/\* surface, with per-page and per-action permission gating.
- Website: a public Blazor SSR site presenting static informational content about the forum and its programme. It carries no sign-in, registration or account page, and it stores no personal data.
- Mobile application: a Flutter app for Android and iOS with a persistent 5-tab shell, built on Riverpod, go\_router and Dio.
- Two databases: the physically separated SIMF\_Identity (SimfidentityDbContext) and SIMF\_App (SimfAppDbContext) SQL Server databases, each with its own migration history.
- Shared libraries: SIMF.Common, SIMF.Contracts, SIMF.ApiClient, SIMF.Components.

The feature surface covered is the full bounded-context set:

- Auth/IdentityAccess
- Account/MyArea
- Programme: sessions, themes, speakers, presentations, summaries, recordings
- SeatReservations/Bookings
- Gates
- Exhibition/Exhibitors/Booths
- Sponsors
- News/Media
- Archive
- Notifications
- Networking
- MeetingRequests + BusinessMeetings
- Statistics
- Configuration
- Organisations
- Contacts
- FAQ
- Feedback
- CMS
- AI
- Venue
- Operations
- Auditing/Logs

Later sections cover every one of these contexts. Where full prose is impractical, a section uses an index table that carries every item rather than dropping any.

### 1.2.1 Out of scope

The following are excluded from this design.

- Real-time push. Live notifications and Q&A are delivered over REST and polled by the client on a bounded interval. No server-push transport ships in this build.

- Distributed cache and scale-out. No distributed cache is part of this design, and no scale-out cluster topology is specified.
- Load-test thresholds, and the monitoring and alerting toolchain. Neither is specified in this design and both remain open.
- The geofence arrival and attendance chain, which is a deferred open item.
- A different production live-video provider. Live video is delivered on YouTube with an HLS/MP4 fallback, which is the decision for this build; procuring an alternative provider is out of scope.
- Payment. There is no payment, ticketing, invoicing or finance capability anywhere in the platform. Attendance is free. This is an explicit non-goal.
- Identity federation. There is no ministry directory integration, no single sign-on, no Active Directory, no LDAP and no NAFATH. The users are members of the public and are not ministry personnel, so there is nothing to federate. SIMF runs its own identity store. This is an explicit non-goal.
- Sign language and live translation, which are withdrawn. Two endpoints for them remain in the API and no client calls them. Their removal is an open item.
- Remediation of committed development secrets, which the owner and operations complete before handover. It is not a code deliverable of this design.
- The external systems themselves. Two systems sit outside SIMF and neither is designed here: the video platform, which is an integration endpoint called over a public URL, and the ministry log collector, which receives the audit trail.

The AI model, the mail relay and the object store are not external systems. They run on site in HSA, the security area that carries internal traffic only and has no internet path, and this document describes them like any other server. The data that crosses the system boundary is itemised in the per-service data statement that accompanies the data-flow figure.

### 1.3 Definitions and Acronyms

The terms and acronyms used throughout this document are defined below.

| Term / Acronym         | Definition                                                                                                                             |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| API                    | Application Programming Interface, the FastEndpoints HTTP backend, SIMF.Api                                                            |
| CP                     | Control Panel, the Blazor Server admin application                                                                                     |
| BFF                    | Backend-for-Frontend, the thin server-side endpoint layer in the Control Panel and Website that holds tokens so the browser never does |
| DDD                    | Domain-Driven Design                                                                                                                   |
| JWT                    | JSON Web Token, the RS256 access token issued by the token service                                                                     |
| TOTP                   | Time-based One-Time Password, the authenticator-app second factor used by admins                                                       |
| OTP                    | One-Time Password, the e-mailed code second factor used by visitors                                                                    |
| RBAC                   | Role-Based Access Control                                                                                                              |
| PII                    | Personally Identifiable Information, encrypted at rest via IPiiEncryptor                                                               |
| RTL / LTR              | Right-to-Left and Left-to-Right text direction, for Arabic and English                                                                 |
| NCA                    | National Cybersecurity Authority (Saudi Arabia)                                                                                        |
| SSR                    | Server-Side Rendering, used by the Blazor Website                                                                                      |
| SoT                    | Source of Truth                                                                                                                        |
| ApiResponse / ApiError | The standard response envelope, ApiResponse<T>, and its error object                                                                   |
| PermissionCatalog      | Single source of truth for permission codes in the form Page.Action, reused by the API and the Control Panel                           |
| GridQuery / GridPage   | The admin-grid paging request and response contract in SIMF.Common                                                                     |
| MobileAppRole          | The mobile privilege model: None / Visitor / Staff / Moderator / Exhibitor                                                             |

| Term / Acronym | Definition                                                                                           |
|----------------|------------------------------------------------------------------------------------------------------|
| EF Core        | Entity Framework Core, the code-first ORM                                                            |
| Soft delete    | IsActive-based deactivation via Deactivate(); list queries filter IsActive == true                   |
| Logical FK     | A cross-database user reference stored as a bare Guid, resolved on read, never a database constraint |
| RSNF           | Royal Saudi Naval Forces, the forum's host                                                           |
| SIMF           | Saudi International Maritime Forum, the platform this document designs                               |
| SDD            | Solution Design Document, this document, also referred to as the Low Level Design                    |

## 1.4 References

The delivery is three documents and they are self-contained. Each is listed below with the area it governs. No other document is needed to read, review or accept this design.

| Document                       | Identifier and version                     | Area it governs                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Business Requirements Document | SIMF-BRD-001, version 2.0, English edition | The business objectives, the scope, the stakeholders, the business rules, and the numbered business, functional, non-functional and operational requirement set.                                                                                                                                                                                                                                                                                                                  |
| High Level Design              | SIMF-HLD-005, version 2.0                  | The solution architecture, the security areas and network zones, the integration and data architecture views, the security view, and the sizing and non-functional views. It is the entry point above this document.                                                                                                                                                                                                                                                              |
| Solution Design Document       | SIMF-LLD-004, version 2.0                  | This document, the detailed technical design. The response envelope, the standard headers, the error model, the HTTP status codes and the authentication endpoints are in section 2.1.4. The permission catalogue and the account-state and booking-status enumerations are in section 2.1.5 and section 6. The screen and route inventory for every client is in section 4. The deployment, health and rollback model is in section 7. Requirement traceability is in section 8. |

Requirement traceability is a numbered section of this document, section 8. It is not an annex. Section 8 lists every requirement identifier from the Business Requirements Document, the section here that designs it, and the verification method for it.

## 2. Process View

SIMF at runtime: the deployed components that talk to each other, what a request passes through on its way to an endpoint, and how the four highest-value business flows execute end to end. The static structure, the projects, the layering and the schema, belongs to later sections.

### 2.1 Process Interaction

#### 2.1.1 Participating components

At runtime SIMF is a small set of long-running processes over HTTP, one pair of databases, and the scheduled jobs the API runs in its own process. The one API is the sole writer to either database; four clients call it. The mobile application reaches it through the mobile edge, SIMF.MobileEdge, the mobile presentation tier the shipped app addresses, forwarding the app surface inward. The public website and the Control Panel call it directly. The on-site badge desk, SIMF.BadgeDesk, a WinForms application, posts its offline badge batches to POST /api/v1/admin/offline/batch. Every host also ships structured logs to the ministry log collector over syslog on TCP 6514.

| Component | Kind | Talks to | Protocol or mechanism |
|-----------|------|----------|-----------------------|
|-----------|------|----------|-----------------------|

| Component                                  | Kind                       | Talks to                                | Protocol or mechanism |
|--------------------------------------------|----------------------------|-----------------------------------------|-----------------------|
| Mobile app (Flutter)                       | Client                     | API                                     | REST over HTTPS       |
| Control Panel (Blazor Server)              | Client                     | API                                     | REST via typed client |
| Website (Blazor SSR with islands)          | Client                     | API                                     | REST via typed client |
| SIMF.Api (FastEndpoints host)              | Server                     | Both databases                          | EF Core               |
| SimfIdentityDbContext and SimfAppDbContext | Persistence                | Two SQL Server databases                | EF Core               |
| Scheduled jobs                             | In-process hosted services | App database, Identity database, e-mail | EF Core and SMTP      |

### 2.1.2 Client-to-API communication

- All four clients call one REST API under the route prefix `api/v1`. Routes split by convention into an app surface, `/api/v1/app/*`, and an admin surface, `/api/v1/admin/*`, exposed as two OpenAPI documents.
- Every response uses the `ApiResult<T>` envelope, fields `Success`, `Data`, `Error` and `Meta`. `ApiCallResult<T>` pairs it with the HTTP status, so a client can react to 401 and 403.
- The mobile app uses one `SimfApiClient`, built on Dio, whose interceptors attach `X-App-Key`, `X-Device-Type`, `Accept-Language` and `Authorization: Bearer` with the access token. On a 401 it refreshes once through `POST /app/auth/refresh` and replays the request; concurrent 401 responses share one in-flight refresh.
- The Control Panel and the Website are server-side Blazor applications using the typed `SIMF.ApiClient`: `SimfAuthClient`, `SimfAccountClient` and `SimfAdminClient` for the Control Panel, `SimfPublicClient` for the Website. Both call server to server, so the browser never holds a raw bearer token.
- Traffic between the presentation and application tiers is TLS 1.2 or higher, with mutual TLS between them. A host without a valid client certificate is refused before any token is examined.
- Under the backend-for-frontend pattern the Control Panel endpoint layer runs the credential and second-factor steps, stores the tokens in an encrypted cookie, holds the access token per circuit in `SimfAuthSession` and forwards it on each call. The Website signs nobody in and holds no bearer token. Network faults map to a failed `ApiResult` rather than being thrown.

### 2.1.3 API-to-database and background communication

- The API owns two physically separated databases through two EF Core contexts: `SimfIdentityDbContext` for users, roles, permissions and tokens, `SimfAppDbContext` for all business data. A reference crossing the boundary is a bare Guid, a logical foreign key resolved on read, never a cross-database join.
- Scheduled jobs run in process as hosted services inside the API and take no part in the request path: `DormantAccountSweepService`, `RegistrationGateAutoCloseWorker`, `SessionReminderWorker`, `ReservationNoShowReleaseWorker` (the per-minute no-show seat release of section 2.2.3), the rating-prompt, meeting-reminder and notification-broadcast workers, the retention and hall-attendance closeout sweeps, and the e-mail dispatcher. There is no separate worker server.
- Every job runs under a single-instance lease on the database, so across four API nodes each job runs exactly once. A node losing the lease stands its workers down.

### 2.1.4 The request/middleware pipeline



# SIMF API request and response pipeline

Every request descends the stack and every response ascends it. The stage order is the order section 2.1.4 lists.

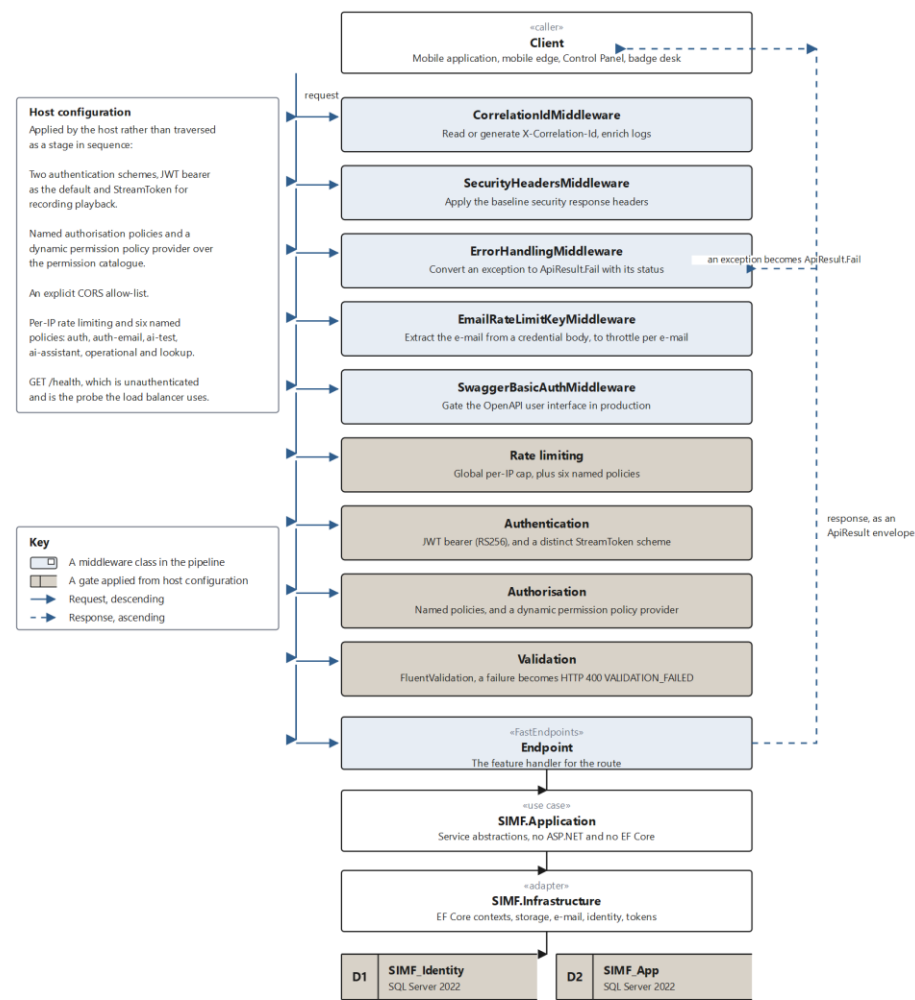


Figure 1. The API request and response pipeline. Every request descends the stage stack and every response ascends it. The stage order is the order this section lists.

Every API request passes a fixed middleware pipeline before reaching an endpoint. Host configuration adds JWT bearer authentication alongside a distinct StreamToken scheme, the named and dynamic permission authorisation policies, an explicit CORS allow-list, per-IP and named rate limiting, and the health endpoint GET /health.

| Class                       | Responsibility                                                                |
|-----------------------------|-------------------------------------------------------------------------------|
| CorrelationIdMiddleware     | Reads or generates X-Correlation-Id; enriches the logs with it                |
| SecurityHeadersMiddleware   | Applies the baseline security response headers                                |
| ErrorHandlingMiddleware     | Converts an exception into ApiResult.Fail with the correct HTTP status        |
| EmailRateLimitKeyMiddleware | Extracts the e-mail address from a credential body for per-address throttling |
| SwaggerBasicAuthMiddleware  | Gates the OpenAPI user interface in production                                |

The pipeline decides the response code. Authentication failure returns 401, covering an invalid token, an expired token and a security stamp that no longer matches. A failed permission check returns 403. A write a uniqueness constraint rejects, such as a second reservation for an occupied seat, returns a conflict. Every response carries the ApiResult envelope, so a client reads outcome and status together through ApiCallResult<T>.

### 2.1.5 Authentication and permission flow

- An endpoint declares its own authorisation in Configure(). AllowAnonymous() covers the public read endpoints, the authentication endpoints that run before a caller can hold a bearer token, the single-use speaker action-token endpoint, the public contact form POST /app/contact-inquiry, and the two AI endpoints POST /app/ai/faq and POST /app/ai/translate. An admin action instead declares Policies(...) naming a permission policy together with RequireApprovedAccount. The speaker action-token, contact-form and AI endpoints are rate-limited under the auth policy.
- Tokens are RS256 JSON Web Tokens issued by JwtTokenService, carrying sub, email, jti, security\_stamp, account\_state, user\_type and mobile\_app\_role, one claim per role, and one perm claim per permission code. The Administrator wildcard collapses to a single perm claim holding \*.
- Validation pins the algorithm to RS256 and checks issuer, audience, lifetime and signature. A security-stamp check makes revoking a role or disabling an account take effect immediately rather than at token expiry.
- The signing key is asymmetric: the API holds the private half inside HSA and the verifiers only the public half, so no presentation-tier host can mint a token.
- A successful second factor issues an access token and a hashed, rotating refresh token, under an absolute session cap of 24 hours, after which the holder authenticates again.
- PermissionCatalog is the single source of truth. PermissionPolicyProvider materialises a perm:<code> policy on demand; PermissionAuthorizationHandler passes the caller whose perm claims hold the code or the wildcard. The Control Panel reuses the catalogue, so both enforce identical codes.

## 2.2 Process Flow

The four walkthroughs below are the primary end-to-end business processes, each carrying its decision points and exception paths and tracing a path through the data flow shown next.

Three services sit outside the application tier and receive data from it. The video platform is called over a public URL; as an integration endpoint rather than a component of this solution, it appears in the data flow view above and deliberately not in the component view. What crosses the system boundary to each is stated below.

Video platform, YouTube. The media team publishes the live session, the summary video and the public video library on the forum's YouTube channel, surfaced in SIMF as links, so the library needs no storage, encoding or bandwidth on ministry servers. SIMF uploads nothing and relays no stream; no attendee record or personal data accompanies either. The platform receives video content and a public video identifier. Session recordings are the exception: an administrator uploads the recording file, the API stores it in the file store and streams its bytes to the app against a stream token.

AI service, on site in HSA. The SITE-hosted GPT OSS 120B model runs on the on-site LLM server, reached over an OpenAI-compatible API. It receives session subtitle text for summarisation, attendee questions submitted during a session for filtering before the moderator queue, and FAQ and support text for the assistant. The unit of work is the content itself: no attendee identity, account record or operational data accompanies a call, and nothing leaves the ministry network.

E-mail relay, on site in HSA. It receives notifications only: the account, booking, session and announcement messages the platform raises, each addressed to its recipient. No session content, and nothing from the AI or video paths.

One external service needs a network exception: the Control Panel reaches the YouTube caption API over outbound HTTPS on port 443 and requires an approved firewall exception from SSA to those hosts. Until it is approved, an administrator pastes or uploads the subtitle. The AI runs on site and needs none.

Caption language and Arabic. The YouTube caption API returns the caption tracks a published video carries, and for a live broadcast serves English only. SIMF therefore produces no live Arabic transcription and no live translation, and claims none. The Arabic path runs after the session: the published caption track is imported, or an administrator pastes or uploads the transcript, and the on-site GPT OSS 120B model translates and summarises it. The Scientific Committee reviews and approves the result before attendees see it, so the end-to-end time is the minutes or hours of that review, not a real-time figure.

Sign language and live translation are withdrawn from the delivered scope. Two endpoints from that work remain in the API, called by no client; their removal is an open item.

### 2.2.1 Visitor sign-in with e-mail OTP

1. The visitor submits e-mail and password in the app, which calls POST /app/auth/sign-in. The route is anonymous and rate-limited under the auth and auth-email policies.

2. The API verifies the credential. Decision point: is the credential valid? Exception path: an invalid credential returns an ApiResult.Fail carrying an invalid-credential error and the flow stops.

3. On a valid credential the API issues a one-time code, an AccountCode with purpose SignInOtp, and queues an e-mail that EmailBackgroundService drains and sends over SMTP. The API responds that a second factor is required.

4. The visitor enters the code and the app calls POST /app/auth/verify-otp.

5. Decision point: is the code correct and unexpired? The API checks it against ExpiresAt, ConsumedAt and AttemptCount. Exception path: a wrong or expired code returns a failure and increments the attempt count, and repeated failures are throttled.

6. On success the API issues an RS256 access token carrying the perm and role claims, and a hashed, rotating refresh token, under an absolute session cap of 24 hours.

7. The app stores both tokens, and HeadersInterceptor attaches the bearer token on subsequent calls. A later 401 triggers a single-flight refresh through POST /app/auth/refresh.

Administrator variant. The same flow with a different second factor: a TOTP code, with a single-use recovery code as the fallback, rather than an e-mailed one-time code.

### 2.2.2 Admin permission-gated action

1. An administrator triggers a Control Panel action, for example listing visitors. The Control Panel calls the API server to server through SimfAdminClient, forwarding the access token held for that circuit.

2. The request passes the middleware pipeline, the correlation id, the security headers and the error handling, and reaches JWT authentication.

3. Decision point: is the token valid and the security stamp current? Exception path: an invalid or expired token, or a stamp no longer matching because a role was revoked or the account disabled, fails authentication and returns 401.

4. The endpoint declares Policies(PermissionCatalog.PolicyFor(<code>), RequireApprovedAccount). PermissionPolicyProvider materialises the perm:<code> policy and PermissionAuthorizationHandler checks the caller's perm claims.

5. Decision point: does the caller hold the code or the wildcard, and is the account approved? Exception path: the API returns 403 and the Control Panel surfaces the gate. The Control Panel also gates the page with RequirePermission, the navigation item and the action button, enforcing the gate on both sides.

6. On a pass, HandleAsync() runs the use case. Mutating operations are stamped and row-audited by the SaveChanges interceptors, and the response returns as ApiResult<T>.

### 2.2.3 Seat reservation, auto-confirmed and confirmed at check-in

1. An approved visitor calls POST /app/sessions/{sessionId}/seats/reserve. The endpoint validates the request and resolves the caller from the JWT under RequireApprovedAccount.

2. The seat-reservation service loads the Session with its Hall and HallSeatLayout, and reads the SeatSelectionMode.

3. Decision point: is the seat free? A filtered unique index permits one active reservation per seat per session. Exception path: a taken seat fails the write and the API returns a conflict.

4. The service creates a SeatReservation with Kind UserBooking and Status Approved on create, the reservation-only auto-confirm. The audit interceptors stamp and row-audit the insert.

5. The reservation is confirmed immediately and stands as a provisional hold. A per-minute sweep releases any hold whose holder has not checked in by three minutes before the session starts, exempting a reservation made after that deadline and a walk-in hold recorded at the door.

6. At the hall gate a staff member scans the attendee's QR code, which marks the held seat confirmed.

7. The app shows an inline success message on reservation. No booking notification is queued or delivered.

#### 2.2.4 Gate scan, idempotent

1. A staff member scans an attendee badge QR code at a gate, and the app calls POST /app/gates/{gateId}/scans. The route is gated by the staff mobile\_app\_role and by the gate permission policy.

2. Decision point: is this a duplicate scan? The service consults ScanIdempotency; a scan already recorded returns the prior outcome and writes no second row. That idempotence is the core guarantee of the flow.

3. On a first-seen scan the service resolves the attendee and the gate's allowed profile types, held in GateProfileTypeAllow, and evaluates the scan.

4. Decision point: is entry permitted for this attendee at this gate? The outcome becomes a GateScan row in an append-only table with a bigint identity key, storing actor and subject snapshot columns and excluded from row-audit as an immutable log.

5. The API returns the scan outcome, allowed or denied with a reason, as ApiResult<T>, and the staff app shows it. A denial or error is recorded on that same append-only row through ScanOutcome, not raised as an exception that loses the scan.

### 3. System Design

The SIMF use case model: the primary actors, the major use cases of every module at the system boundary, a full use case realisation for five key modules, and an index covering the remainder. Use case identifiers take the form UC-nn, as declared for each feature. A delivered use case with no numbered identifier is marked Not numbered rather than given an invented number.

#### 3.1 Use Case Realization

##### 3.1.1 High Level Use Case

The actor set follows the account and role model. MobileAppRole has five values: None, Visitor, Staff, Moderator and Exhibitor; the mobile application adds Guest for a caller holding no token. Exhibitor gates the booth lead capture screens and the endpoints behind them. Control Panel access is assigned by role only, and the Administrator role carries the wildcard permission code "\*".

| Actor              | Surface                                          | Description                                                                                                                                                                                                                                                                                |
|--------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anonymous / Public | Website, mobile application as Guest             | Unauthenticated visitor browsing public content: programme, speakers, news, archive, delegations and the venue map. Holds no account. The forum is open to the general public and international in attendance, restricted neither to Ministry of Defense personnel nor to Saudi nationals. |
| Visitor / Attendee | Mobile application, Website                      | Self registered and approved end user, UserType = Visitor and MobileAppRole = Visitor. Books seats, asks questions, rates sessions and networks.                                                                                                                                           |
| VIP / Delegate     | Mobile application, managed in the Control Panel | Visitor flagged VIP or VVIP, or carrying IsDelegate with an invited country. Welcome roster maintained in the Control Panel at /admin/visitors/vip and /admin/delegates, with a delegations flag.                                                                                          |
| Moderator          | Mobile application,                              | Runs the per session question desk, for assigned sessions only.                                                                                                                                                                                                                            |

| Actor                     | Surface                                   | Description                                                                                                                                                                                     |
|---------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | MobileAppRole = Moderator                 |                                                                                                                                                                                                 |
| Gate operator / Staff     | Mobile application, MobileAppRole = Staff | Scans badges at venue gates and hall doors; runs the on site registration desk.                                                                                                                 |
| Administrator / Organiser | Control Panel                             | UserType = Admin. Staffed by Royal Saudi Naval Forces personnel and the organising company. Every Control Panel action is gated by a permission code; the account carries a TOTP second factor. |
| Scientific committee      | Control Panel                             | Reviews the programme, the live broadcast and the team stage of the question pipeline.                                                                                                          |
| Public Relations          | Control Panel                             | Approves exhibitors and assigns booths, through AppRoles.PublicRelations. Attendee bookings are confirmed on create, so this role holds no booking approval queue.                              |
| System / scheduled jobs   | Server                                    | Dormant account sweep, registration gate auto close, session reminders, rating prompts and e-mail dispatch.                                                                                     |

The authoritative account model is UserType with ProfileType. Scientific committee and Public Relations are role assignments in the suggested configuration, not fixed system roles.

The system boundary use case summary lists the major use cases of every module. A use case identifier is given where the governing specification names one.

| Module                    | Use case ID  | Name                                | Actors                              | Description                                                                                                                                            |
|---------------------------|--------------|-------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authentication            | UC-01        | Create an account                   | Anonymous                           | Sign up with e-mail and password to reach Registered.                                                                                                  |
| Authentication            | UC-02        | Verify the e-mail                   | Anonymous                           | Submit a 6 digit code to reach EmailVerified; resend supported.                                                                                        |
| Authentication            | UC-04        | Sign in                             | Visitor, Administrator              | Password with a branch on the account state, then an e-mail OTP for a visitor or TOTP for an administrator, then the audience gate for cp, web or app. |
| Authentication            | UC-06        | Reset a password                    | Any actor                           | Forgot and reset through an e-mailed code, invalidating the active sessions.                                                                           |
| Authentication            | Not numbered | Refresh a token and sign out        | Any signed in actor                 | Rotate the refresh token within a 24 hour cap; revoke on sign out.                                                                                     |
| Authentication            | Not numbered | Enrol or reset TOTP                 | Administrator                       | Two slot authenticator enrolment with 10 recovery codes, plus an administrator driven second-factor reset.                                             |
| Registration and Approval | UC-03        | Complete registration               | Visitor, Exhibitor                  | Profile, identity, attachments, venue track and terms, submitted to PendingApproval.                                                                   |
| Registration and Approval | UC-07        | View registration status            | Visitor                             | Stages: data sent, e-mail, security review, activation.                                                                                                |
| Registration and Approval | UC-20        | Review and decide a registration    | Administrator, Scientific committee | Approve, setting the final type, minting the QR identifier and issuing the badge, or reject with a reason. Bulk approval supported.                    |
| Registration and Approval | UC-21        | Approve an exhibitor                | Public Relations                    | One stage approval that also assigns the booth.                                                                                                        |
| Registration and Approval | UC-31        | Open or close registration          | Administrator                       | Manual control, plus auto close at an administrator-set date and time.                                                                                 |
| Registration and Approval | UC-34        | Register on site or reprint a badge | Staff                               | Desk search, then badge reprint or on site registration.                                                                                               |
| Programme                 | UC-23        | Manage sessions                     | Scientific committee,               | Create, read, update and delete sessions; link theme, hall and speakers; set the live flag.                                                            |

| Module                  | Use case ID  | Name                                        | Actors                              | Description                                                                                                                                                                |
|-------------------------|--------------|---------------------------------------------|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |              |                                             | Administrator                       |                                                                                                                                                                            |
| Programme               | UC-24        | Manage halls and seating                    | Administrator                       | Halls, capacity, seat grid and seat layouts.                                                                                                                               |
| Programme               | UC-25        | Manage speakers                             | Administrator                       | Speaker profiles and their presentations.                                                                                                                                  |
| Programme               | UC-08        | Browse the agenda and a session             | Public, Visitor                     | Day filter, search, session detail, add to calendar and reminders.                                                                                                         |
| Bookings and Attendance | UC-09        | Book a session seat                         | Visitor                             | Pick a seat under the no overlap rule, confirmed on create. A sweep releases it three minutes before the session starts if the holder has not checked in at the hall gate. |
| Bookings and Attendance | UC-10        | Cancel a booking                            | Visitor                             | Cancel before the session starts; the seat is released.                                                                                                                    |
| Bookings and Attendance | UC-22        | Approve a booking                           | Public Relations                    | Not used. No attendee booking enters Pending: a reservation is created in Approved and stays there until cancelled or released.                                            |
| Bookings and Attendance | Not numbered | View the seat map and My Seat               | Visitor                             | Seats shown as available, taken or mine, the assigned seat on the session detail.                                                                                          |
| Gates and Access        | UC-11        | View my badge and QR                        | Visitor                             | Badge card and entry QR code.                                                                                                                                              |
| Gates and Access        | UC-33        | Verify a badge at venue entry               | Staff                               | Scan, verify, then record a GateScan row.                                                                                                                                  |
| Gates and Access        | UC-35        | Check in at a hall door                     | Staff, door device                  | A QR scan writes a HallAttendance record. Geofence arrival is deferred, open item G-OI-2.                                                                                  |
| Gates and Access        | UC-12        | Scan another attendee's badge               | Visitor                             | Save a contact from a badge QR code.                                                                                                                                       |
| Exhibition              | UC-18        | Browse exhibitors, booths and the venue map | Public, Visitor                     | Booth directory, sponsors by tier, and a 2D venue map with navigation.                                                                                                     |
| Exhibition              | Not numbered | Capture an exhibitor lead                   | Exhibitor, approved                 | Scan a visitor badge into my visitors.                                                                                                                                     |
| Sponsors                | UC-18        | View sponsors                               | Public, Visitor                     | Sponsor list and sponsor detail, ordered by tier.                                                                                                                          |
| News, Media and Archive | UC-32        | Manage and view media, news and archive     | Administrator manages, public reads | Media Center, news and categories, gallery, media partners, previous editions and their visibility.                                                                        |
| Engagement and Q&A      | UC-13        | Watch a live session                        | Visitor                             | Live stream, caption language, and an optional notification to the session's seat holders.                                                                                 |
| Engagement and Q&A      | UC-14        | Ask a question in a session                 | Visitor                             | Recipient is a speaker or the host. Arrival gated. Phase is Pre or Live.                                                                                                   |
| Engagement and Q&A      | UC-36        | Manage the questions of an assigned session | Moderator                           | Order, hide, put on air and mark answered.                                                                                                                                 |
| Engagement and Q&A      | Not numbered | Rate a session, a day or the event          | Visitor                             | Triggers: day, programme end, session end and live close.                                                                                                                  |
| Networking              | UC-16        | Request a one to one meeting                | Visitor                             | Meeting request, then approval.                                                                                                                                            |
| Networking              | UC-17        | Use the AI                                  | Visitor                             | Grounded on the programme, the FAQ and the                                                                                                                                 |

| Module                | Use case ID  | Name                                  | Actors              | Description                                                                                                                 |
|-----------------------|--------------|---------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
|                       |              | assistant                             |                     | booth list.                                                                                                                 |
| Networking            | Not numbered | Meet people like you                  | Visitor             | Interest based matchmaking, recommending and pushing at a match of 80 percent or higher.                                    |
| Networking            | Not numbered | AI session summary                    | Visitor             | Generated per session summary.                                                                                              |
| Networking            | Not numbered | Accessibility settings                | Visitor             | Text size, high contrast, reduced motion, screen reader assistance and captions.                                            |
| Meetings, B2B and B2C | Not numbered | Schedule a business meeting           | Administrator       | Configure the hall and its tables, reserve over a slot, and schedule two or more parties to Confirmed. Detail in section 5. |
| Meetings, B2B and B2C | Not numbered | Request a meeting in a hall           | Visitor             | Attendee request, then administrator review against hall availability, then a speaker double opt in.                        |
| Notifications         | UC-19        | Receive a notification                | Any signed in actor | One abstraction over the in app and e-mail channels, with an inbox and reminders. Detail in section 5.                      |
| Statistics            | UC-30        | View the statistics dashboard         | Administrator       | Per day figures, overall figures and live attendance. Detail in section 5.                                                  |
| Configuration         | UC-28        | Manage dynamic content and categories | Administrator       | Content blocks, banners, categories, labels, colours and venue tracks. Detail in section 5.                                 |
| Configuration         | UC-31        | Open or close registration            | Administrator       | Registration control. Detail in section 5.                                                                                  |
| Configuration         | Not numbered | View the operation log                | Administrator       | History of every change.                                                                                                    |
| Contacts and Sharing  | Not numbered | Share my contact by QR or vCard       | Visitor             | Consent by action share token with a vCard. Track 2.                                                                        |
| Contacts and Sharing  | Not numbered | Scan and save a contact               | Visitor             | Resolve a shared token into My Contacts. Track 2.                                                                           |

No module is omitted. The per page and per action inventory of the Control Panel, the Website and every mobile screen is section 4.

### 3.1.2 Low Level Use Cases

The five modules below carry full use case tables. Flow steps are numbered to the processing steps, and the error and exception codes named are those the system returns.

Module 1, authentication and sign in. Realised in full: UC-04, sign in with a second factor.

| Field         | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID   | UC-04                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Name          | Sign in                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Actor         | Visitor on the Website or the mobile application; Administrator on the Control Panel.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Preconditions | The account exists and is EmailVerified or beyond. The caller supplies the surface audience: cp, web or app.                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Main Flow     | 1. The actor submits e-mail and password. 2. The system finds the account by e-mail and verifies the password hash. 3. It branches on the account state. 4. Where a second factor applies: TOTP for a user holding an authenticator key, otherwise an e-mail OTP for a visitor, returning mfaRequired with a short lived token. 5. The actor submits the TOTP or OTP code, and the system validates both the code and the mfa or otp token. 6. Audience gate: the surface must match, cp to Admin and web or app to Visitor. 7. It issues an access token |



| Field             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                   | valid for 5 minutes and a rotating refresh token capped at 24 hours. 8. It writes SignIn.Succeeded to the operation log.                                                                                                                                                                                                                                                                                                                                                   |
| Alternative Flows | A1. TwoFactorEnabled = false: tokens are issued on the password step and MfaRequired = false. A2. PendingApproval on web or app: allowed with a limited status view session, refused on cp. A3. The recovery code path replaces TOTP through /auth/verify-recovery-code.                                                                                                                                                                                                   |
| Exception Flows   | Bad credentials: the generic AUTH_INVALID_CREDENTIALS with 401. Unverified e-mail: AUTH_EMAIL_NOT_VERIFIED with 403. Rejected, disabled or locked account: AUTH_ACCOUNT_NOT_APPROVED, AUTH_ACCOUNT_DISABLED or AUTH_ACCOUNT_LOCKED with 403. Wrong second factor: AUTH_TOTP_INVALID; expired mfa token: AUTH_MFA_TOKEN_EXPIRED, both with 400. Wrong surface: 403 with AUTH_WRONG_SURFACE_CP or AUTH_WRONG_SURFACE_WEB. Exceeded rate limit: RATE_LIMIT_EXCEEDED with 429. |
| Postconditions    | A session exists as a token pair. The access token carries the account state, the user type, the permission codes and mobile_app_role. Audit rows are written.                                                                                                                                                                                                                                                                                                             |

Companion use cases in the same module: UC-01 sign up, UC-02 verify the e-mail with resend, UC-06 password reset, token refresh and sign out, and TOTP enrolment, recovery and administrator reset.

Module 2, registration and approval. Realised in full: UC-03 and UC-20, complete a registration and decide it.

| Field             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID       | UC-03 to register, UC-20 to review and decide.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Name              | Complete the registration profile, and the organiser decision on it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Actor             | Visitor or Exhibitor registers; Administrator or a reviewer role decides.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Preconditions     | The account is EmailVerified and registration is open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Main Flow         | 1. The user picks a registration type: Visitor, Other or Exhibitor. 2. Personal data: the four part Arabic name, the English or passport name, nationality, date of birth and place of birth. 3. Identity data: a Saudi national identity number of 10 digits starting with 1, or for a non Saudi a passport number or an Iqama number of 10 digits starting with 2. 4. Contact numbers, at least one required: a Saudi mobile number must be 05XXXXXXX or +9665XXXXXXX, an international one E.164. 5. An identity document image is uploaded, encrypted at rest with AES-GCM and passed through a magic byte gate. 6. Identity photo verification, with an alternative path for the women exception. 7. Exhibitor branch: organisation data and companions. 8. A venue track is selected. 9. The user consents to the terms and submits: a RegistrationRequest is created, the account moves to PendingApproval, a confirmation message is sent and an operation log entry written. 10. The reviewer opens the requests queue and reads the data and attachments. 11. On approval the reviewer sets the final user type, the account moves to Approved, a 12 character Crockford QR identifier is minted at that transition, a Badge is issued with a SIMF-2026-xxxx number and a colour, and the user is notified. |
| Alternative Flows | A1. Exhibitor approval, UC-21: Public Relations approves and assigns the booth in one step. A2. Bulk approval through select all. A3. On site registration or badge reprint at the desk, UC-34. A4. The user proposes a ProfileType at sign up, and the administrator assignment wins.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Exception Flows   | A missing mandatory field, or absent consent to the terms, returns VALIDATION_FAILED and blocks the submission. A rejection requires a reason of 10 to 500 characters and moves the account to Rejected. Approving or rejecting a user who is not pending returns 409 with ADMIN_USER_NOT_PENDING. An unknown nationality code returns PROFILE_NATIONALITY_UNKNOWN. A closed registration refuses the submission with a clear message.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Postconditions    | The account is Approved with a QR identifier and a badge, or Rejected with                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Field | Value                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------|
|       | the reason recorded. Audit rows such as Admin.VisitorApproved and Admin.VisitorRejected carry the actor and the subject. |

Companion use cases in the same module: UC-07 registration status tracking, UC-31 open and close registration, internal user onboarding with TOTP enrolment, and the bulk administrator actions.

Module 3, badge and access control at the gate. Realised in full: UC-33 and UC-35, the gate scan at venue entry and at hall arrival.

| Field                      | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                | UC-33 for venue entry, UC-35 for hall check in.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Name                       | Scan a badge at a gate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Actor                      | Gate operator or staff member, or a fixed door device.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Preconditions              | The operator is authenticated, holds Gates Operate and is assigned to the gate. The gate is IsActive.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Main Flow (13-step engine) | 1. The caller is authenticated. 2. Holds Gates Operate and is an active assignee. 3. IQrResolver resolves the QR code to a UserProfile. 4. The idempotency key is checked; a replay returns the earlier outcome. 5. The gate is active. 6 to 8. The holder is Approved, not disabled and not locked. 9. The holder's ProfileType is active. 9.5. Edition check: a badge issued for a closed edition is denied with OutsideTimeWindow; a profile with no recorded edition is left alone. 10. Direction resolves to In, Out or Both from the last allowed scan at this gate. 11. The GateProfileTypeAllow allow list is checked: an empty list is a general pass, a filtered empty result denies as the safe default. 11.5. Session hall check: on a check in at a hall door the attendee must be registered for the session behind that door, else the scan is denied with BookingRequiredMissing. A departure is never blocked, and walk in mode relaxes this rule alone. 12. A 5 second duplicate window absorbs repeats, keyed on GateId and UserProfileId. 13. A GateScan row is persisted with Outcome = Allowed, the resolved direction and the server clock; a denial persists Outcome = Denied with a DenialReasonCode and emits one OperationLog row of GateScanDenied. |
| Alternative Flows          | A1. Hall arrival, UC-35: a QR scan at the door records a HallAttendance enter with Method = QrScan. The GPS geofence method is deferred, open item G-OI-2. Both means firing update one HallAttendance row, not two. A2. Attendee to attendee scan, UC-12: a verified QR code becomes a SavedContact. A3. Currently inside derives from the most recent allowed scan across all gates.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Exception Flows            | QR_UNKNOWN, GATE_INACTIVE_AT_SCAN, HOLDER_NOT_APPROVED, HOLDER_DISABLED, HOLDER_LOCKED, PROFILE_TYPE_INACTIVE and PROFILE_TYPE_NOT_ALLOWED are recorded. IDEMPOTENCY_KEY_CONFLICT returns 409. Failure rate circuit: 10 or more system faults at one gate within 60 seconds open the circuit for 5 minutes, and further scans answer 429 with GATE_FAILURE_CIRCUIT_OPEN. The denials listed above do not feed the circuit.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Postconditions             | A GateScan row is appended, append only by application convention and excluded from RowAudit. Attendance, statistics and the Q&A arrival gate are fed from it, as section 5.5 sets out.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Companion use case: UC-11, view the badge and the QR code, in section 5.3. The GPS geofence hall arrival chain is a deferred programme item; its open points are the retention period and the geofence held on Hall.

Module 4, bookings and attendance. Realised in full: UC-09, reserve a session seat. The reservation is confirmed on create and the gate check in confirms the held seat. The UC-22 approve and reject step is not used.

| Field             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID       | UC-09 to book, UC-10 to cancel, UC-22 to approve.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Name              | Reserve a seat, confirmed on create and held until the gate check in.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Actor             | Visitor with an Approved account. The reviewer approval of UC-22 is not used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Preconditions     | The attendee is Approved. The session is open for booking and a seat is available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Main Flow         | 1. From the session detail the attendee opens the hall seat map, each seat shown as available, held or taken. 2. Selects an available seat. 3. The system checks the no overlap rule, no Pending or Approved booking for a session overlapping in time, and confirms the seat is still free. 4. Creates a SeatReservation in Approved and holds the seat; a filtered unique index allows one active reservation per seat per session. 5. The attendee sees an inline success message. 6. At the hall gate a staff QR scan marks the held seat confirmed. A pre start sweep releases any hold not checked in, three minutes before the session starts. No reviewer queue and no BookingConfirmed notification stand on this path. |
| Alternative Flows | A1. No attendee booking enters Pending, so no review queue stands on this path. A2. A hold is released by the pre start sweep when the attendee has not checked in, and by cancellation. A3. Cancel, UC-10: a Pending or Approved booking cancelled before the session starts moves to Cancelled and the seat is released. A4. Random and open seating reservation kinds are available through SeatReservationKind.                                                                                                                                                                                                                                                                                                              |
| Exception Flows   | An overlap is blocked with an explanation. A seat taken in the meantime raises a constraint violation, handled with an invitation to choose another seat. A full session offers no seats. A cancellation after the session has started is refused. A rejection without a reason is blocked.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Postconditions    | The booking is in a terminal or held state and the seat map is consistent. Attendance is later read from HallAttendance. A session that started with no attendance raises an event for Notifications.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

Module 5, engagement and Q&A. Realised in full: UC-14 and UC-36, ask a question and moderate it.

| Field             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID       | UC-14 to ask, UC-36 to moderate, UC-13 to watch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Name              | Session question, a three stage pipeline.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Actor             | Visitor asks. The scientific committee is the team stage. The moderator runs the per session desk.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Preconditions     | The session exists, is active and has not ended, and the attendee is Approved. Once the session is live, a hall carrying a geofence also requires a HallAttendance enter record for that session.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Main Flow         | 1. The attendee opens the ask card and picks a recipient, a speaker or the host. 2. On submission the backend sets Phase to Pre or Live from the session start time and records the question as Pending. 3. Stage 1: an AI advisory filter sets AiFilterVerdict; it is configuration gated, defaults to a stub, and never hides a question by itself. 4. Stage 2: the scientific committee team reviews the question in the Control Panel at /admin/question-queue and approves, hides or escalates it to a role. 5. Stage 3: the assigned moderator runs the desk in the mobile application for that session alone: view, order, hide, put on air and mark answered. Updates stream live. |
| Alternative Flows | A1. Pre session mode: while the session is upcoming the same ask card carries a distinct ask before the session label, and the question flows through the identical pipeline with Phase = Pre. A2. Watch live, UC-13: the stream, the caption language, and an optional notification to the session's seat holders. A3. Ratings are triggered at session end, at hall departure, at live close, and at the end of a day or of the programme.                                                                                                                                                                                                                                               |
| Exception Flows   | Once the session is live and its hall carries a geofence, an attendee with no hall arrival record is not offered the composer, by design. A hall with no geofence still accepts the question. Outside the window, questions are closed. A moderator opening an unassigned session gets no access. If the AI provider is unavailable the advisory result is ai-unavailable and the pipeline is unaffected.                                                                                                                                                                                                                                                                                  |
| Postconditions    | A SessionQuestion moves from Pending to Approved or to Hidden. An approved question is pushed on stage through IsPushed. Moderator and administrator actions are written to the operation log.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

The remaining modules are indexed below. Attendee comments on a session are not delivered: no entity, no table, no endpoint and no Control Panel page implements them.

| Module                      | Requirements delivered                                                 | Key use cases, with actor                                                                                                                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programme                   | FR-401, FR-402, FR-403, FR-404, FR-405, FR-406, FR-407, FR-408, FR-409 | UC-23 manage sessions; UC-24 manage halls and seating; UC-25 manage speakers and presentations (Administrator, Scientific committee); UC-08 browse the agenda with search, add to calendar and reminders (Public, Visitor).                                                    |
| Exhibition                  | FR-601, FR-602, FR-603, FR-605                                         | UC-18 browse booths, the venue map and navigation (Public, Visitor); manage the booth record after approval (Administrator); exhibitor lead scan (Exhibitor, approved).                                                                                                        |
| Sponsors                    | FR-604                                                                 | UC-18 view sponsors by tier (Public, Visitor); manage sponsors (Administrator).                                                                                                                                                                                                |
| News, Media and Archive     | FR-1001, FR-1002, FR-1003, FR-1004, FR-1005, FR-1006                   | UC-32 manage and read the Media Center, news and categories, gallery, media partners, and previous editions with their visibility (Administrator manages, public reads).                                                                                                       |
| Networking and Cognitive AI | FR-801, FR-802, FR-803, FR-805, FR-806, FR-808                         | UC-16 request a one to one meeting, taken to Public Relations approval (Visitor); UC-17 AI assistant grounded on the programme, the FAQ and the booth list; meet people like you matchmaking, recommending and pushing at a match of 80 percent or higher; AI session summary. |
| Meetings, B2B and B2C       | FR-804, FR-902                                                         | Configure the hall and its tables, reserve over a slot, and schedule two or more parties to Confirmed (Administrator, built); attendee meeting in hall request, then administrator review against hall availability, then speaker double opt in (Visitor, build ready).        |
| Notifications               | FR-901, FR-902, FR-903, FR-904                                         | UC-19 receive a notification across in app, e-mail, SMS and WhatsApp; the in app inbox; reminders; the channel mix held as configuration (any signed in actor; the SMS and WhatsApp providers are deferred).                                                                   |
| Statistics and Dashboards   | FR-1101, FR-1102, FR-1103, FR-1104                                     | UC-30 view the per day, overall, live attendance and GPS presence dashboards (Administrator; the figure list is an open item).                                                                                                                                                 |
| Configuration               | FR-1201, FR-1202, FR-1203, FR-1204, FR-1205, FR-1206, FR-1207, FR-1208 | UC-28 manage content blocks, categories, labels, colours and venue tracks (Administrator); UC-31 open and close registration; view the operation log.                                                                                                                          |
| Contacts and Sharing        | FR-304                                                                 | Share my contact by QR code or vCard, on consent by action; scan and resolve a token, then save it to My Contacts (Visitor); the shared Contact directory referenced by organisation entities (Administrator curated).                                                         |

## 4. User Interfaces

SIMF serves three clients over one backend API: the Control Panel for internal administration (Blazor Server), the public Website (Blazor SSR, static public content) and the mobile application (Flutter, on Android and iOS). This section inventories the screens and routes of all three as delivered, with their audience and purpose. It carries no wireframes and no per-page field, validation or permission detail, which section 5 holds.

### 4.1 Control Panel screens

The Control Panel is a Blazor Server application, rendered interactive-server with prerendering disabled, served under `/admin/*` and authenticated by cookie. Every list page uses the shared `SimfDataGrid`, which supplies the filter, select-all, row checkbox and per-row icon actions over the CRUD framework of

CrudShell, CrudPageFrame, CrudDialogFrame and CrudFormBase. Dialog or full page is a per-page user preference, held as CpPreferences in local storage.

Navigation is organised into 14 permission-gated groups: Overview, People, Access Control, Programme, Scientific Committee, Exhibition, Engagement, Knowledge, Content, Public Relations, Gates, Reference Data, System and Reports. IStringLocalizer<Strings> localises, with dir set from the current UI culture, so the interface is bilingual Arabic and English with right-to-left layout. Colour and typography come from theme.tokens.css.

Every /admin/\* page and action is gated by a PermissionCatalog code, the nested Page.Action constants described in section 2.1.5. Build-failing tests reject an ungated page or endpoint. Administrator in the Audience column names the baseline role holding the page's permission.

The Control Panel serves 113 routes. The catalogue below lists the pages reached from the main navigation, grouped by functional area.

| Area                | Route                        | Screen name           | Audience              | Purpose                                                                                                                                                                 |
|---------------------|------------------------------|-----------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overview            | /                            | Dashboard             | Any signed-in CP user | Landing and overview dashboard                                                                                                                                          |
| People and accounts | /admin/admins                | Admins                | Administrator         | Manage administrator accounts                                                                                                                                           |
| People and accounts | /admin/admins/pending        | Pending admins        | Administrator         | Approve or reject pending admins                                                                                                                                        |
| People and accounts | /admin/others                | Other users           | Administrator         | Accounts neither visitor nor admin                                                                                                                                      |
| People and accounts | /admin/others/pending        | Pending others        | Administrator         | Approve or reject pending others                                                                                                                                        |
| People and accounts | /admin/visitors              | Visitors              | Administrator         | Manage visitor accounts                                                                                                                                                 |
| People and accounts | /admin/visitors/pending      | Pending visitors      | Administrator         | Review and approve visitors: all-data view, tier set on approval, photo viewer                                                                                          |
| People and accounts | /admin/visitors/vip          | VIP registration      | Administrator         | VVIP and VIP registration: the external Mawj system welcome-message fields (Mawj system id, honorific, preferred language) and the VIP photo. Creates a pending account |
| People and accounts | /admin/visitors/vip/export   | VIP export            | Administrator         | VVIP and VIP welcome roster, as JSON, CSV or Excel                                                                                                                      |
| People and accounts | /admin/delegates             | Delegates             | Administrator         | Visitors carrying IsDelegate and an invited country. Single add and bulk badge generation                                                                               |
| People and accounts | /admin/attendees             | Attendees             | Administrator         | Attendee directory                                                                                                                                                      |
| People and accounts | /admin/print-bag             | Print bag             | Administrator         | Badge and print-bag queue                                                                                                                                               |
| People and accounts | /admin/interests             | Interests             | Administrator         | Interest lookup                                                                                                                                                         |
| People and accounts | /admin/profile-types/visitor | Visitor profile types | Administrator         | Visitor profile types                                                                                                                                                   |
| People and accounts | /admin/profile-types/other   | Other profile types   | Administrator         | Non-visitor profile types                                                                                                                                               |
| People and accounts | /admin/organisations         | Organisations         | Administrator         | Organisation lookup                                                                                                                                                     |

| Area                   | Route                           | Screen name              | Audience      | Purpose                                                                                                                                                                                          |
|------------------------|---------------------------------|--------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| People and accounts    | /admin/regions                  | Regions                  | Administrator | Region lookup                                                                                                                                                                                    |
| People and accounts    | /admin/countries                | Countries                | Administrator | Country lookup, with invited flag, delegation head and dates                                                                                                                                     |
| People and accounts    | /admin/vips                     | VIPs                     | Administrator | VIP directory                                                                                                                                                                                    |
| People and accounts    | /admin/invitations              | Invitations              | Administrator | Manage invitations                                                                                                                                                                               |
| People and accounts    | /admin/reset-2fa                | Reset 2FA                | Administrator | Reset a user's second factor                                                                                                                                                                     |
| People and accounts    | /admin/roles                    | Roles                    | Administrator | Manage roles                                                                                                                                                                                     |
| People and accounts    | /admin/roles/{id}/permissions   | Role permissions         | Administrator | Assign permission codes to a role                                                                                                                                                                |
| Programme and sessions | /admin/themes                   | Themes                   | Administrator | Programme themes                                                                                                                                                                                 |
| Programme and sessions | /admin/halls                    | Halls                    | Administrator | Halls, with capacity and seat mode                                                                                                                                                               |
| Programme and sessions | /admin/halls/seat-layouts       | Seat layouts             | Administrator | Per-hall seat layout editor                                                                                                                                                                      |
| Programme and sessions | /admin/speakers                 | Speakers                 | Administrator | Manage speakers                                                                                                                                                                                  |
| Programme and sessions | /admin/speaker-presentations    | Speaker presentations    | Administrator | Speaker presentation files                                                                                                                                                                       |
| Programme and sessions | /admin/sessions                 | Sessions                 | Administrator | Manage sessions                                                                                                                                                                                  |
| Programme and sessions | /admin/sessions/seat-plans      | Session seat plans       | Administrator | Per-session seat plan                                                                                                                                                                            |
| Programme and sessions | /admin/session-categories       | Session categories       | Administrator | Session category lookup                                                                                                                                                                          |
| Programme and sessions | /admin/programme-days           | Programme days           | Administrator | Programme day definitions                                                                                                                                                                        |
| Programme and sessions | /admin/session-moderators       | Session moderators       | Administrator | Assign session moderators                                                                                                                                                                        |
| Programme and sessions | /admin/programme/timeline       | Programme timeline       | Administrator | Programme timeline view                                                                                                                                                                          |
| Programme and sessions | /admin/bookings                 | Bookings                 | Administrator | Read-only monitor of active seat reservations, confirmed and still held, with export. No approval step, no approve or reject control. Bookings.View and Bookings.Export are its only permissions |
| Programme and sessions | /admin/speaker-meeting-requests | Speaker meeting requests | Administrator | Review speaker meeting requests                                                                                                                                                                  |
| Programme and sessions | /admin/speaker-availability     | Speaker availability     | Administrator | Speaker availability windows, generating the free slots offered when a visitor requests a speaker meeting                                                                                        |
| Programme and sessions | /admin/hall-availability        | Hall availability        | Administrator | Team-defined hall meeting-time windows that become free slots                                                                                                                                    |
| Programme and sessions | /admin/delegation-meetings      | Delegation meetings      | Administrator | Review desk for delegation-to-delegation meetings                                                                                                                                                |
| Programme              | /admin/document-requests        | Document                 | Administrator | Participation-document request                                                                                                                                                                   |

| Area                           | Route                        | Screen name        | Audience                  | Purpose                                                                 |
|--------------------------------|------------------------------|--------------------|---------------------------|-------------------------------------------------------------------------|
| and sessions                   |                              | requests           |                           | review, accepted or rejected with a note                                |
| Programme and sessions         | /admin/badge-requests        | Badge requests     | Administrator             | Badge-update request review. Accepting applies the title to the profile |
| Programme and sessions         | /admin/meeting-tables        | Meeting tables     | Administrator             | Business-meeting tables                                                 |
| Programme and sessions         | /admin/business-meetings     | Business meetings  | Administrator             | Manage business meetings                                                |
| Engagement, Q&A and attendance | /admin/question-queue        | Question queue     | Administrator             | Audience question moderation queue                                      |
| Engagement, Q&A and attendance | /sessions/{id}/moderate      | Session moderation | Session moderator         | Live moderation of a session's questions                                |
| Engagement, Q&A and attendance | /admin/ratings               | Ratings            | Administrator             | Rating responses and KPIs                                               |
| Engagement, Q&A and attendance | /admin/rating-config         | Rating config      | Administrator             | Dynamic rating-form configuration                                       |
| Engagement, Q&A and attendance | /admin/session-summaries     | Session summaries  | Administrator             | Author and publish session summaries                                    |
| Engagement, Q&A and attendance | /admin/hall-arrivals         | Hall arrivals      | Administrator or operator | Hall arrival tracking                                                   |
| Exhibition                     | /admin/exhibitors            | Exhibitors         | Administrator             | Manage exhibitors                                                       |
| Exhibition                     | /admin/booths                | Booths             | Administrator             | Booths, with officer, company and map coordinates                       |
| Exhibition                     | /admin/sponsors              | Sponsors           | Administrator             | Sponsors and their tier                                                 |
| Exhibition                     | /admin/media-partners        | Media partners     | Administrator             | Manage media partners                                                   |
| Exhibition                     | /admin/venue-map             | Venue map          | Administrator             | 2D venue-map nodes                                                      |
| Content and media              | /admin/news                  | News               | Administrator             | Manage news items                                                       |
| Content and media              | /admin/media                 | Media              | Administrator             | Media gallery items                                                     |
| Content and media              | /admin/archive               | Archive            | Administrator             | Past-edition archive                                                    |
| Content and media              | /admin/media-library         | Media library      | Administrator             | Unified image-asset library                                             |
| Content and media              | /admin/banners               | Banners            | Administrator             | Site banners                                                            |
| Content and media              | /admin/content-blocks        | Content blocks     | Administrator             | CMS content blocks                                                      |
| Knowledge and AI               | /admin/faq                   | FAQ                | Administrator             | FAQ groups and entries                                                  |
| Knowledge and AI               | /admin/ai                    | AI dashboard       | Administrator             | AI overview dashboard                                                   |
| Knowledge and AI               | /admin/ai/services           | AI services        | Administrator             | AI service list                                                         |
| Knowledge and AI               | /admin/ai/services/{feature} | AI service detail  | Administrator             | Per-feature AI service configuration                                    |
| Knowledge and AI               | /admin/ai/prompts            | AI prompts         | Administrator             | Prompt management                                                       |

| Area                      | Route                       | Screen name          | Audience      | Purpose                                                         |
|---------------------------|-----------------------------|----------------------|---------------|-----------------------------------------------------------------|
| Knowledge and AI          | /admin/ai/invocations       | AI invocations       | Administrator | AI invocation telemetry                                         |
| Access control and system | /admin/gates                | Gates                | Administrator | Manage gates                                                    |
| Access control and system | /admin/gates/operator       | Gate operator        | Gate operator | Entry and exit scanning console                                 |
| Access control and system | /admin/gates/dashboard      | Gate dashboard       | Administrator | Gate activity dashboard                                         |
| Access control and system | /admin/configuration        | Configuration        | Administrator | System settings, as key and value                               |
| Access control and system | /admin/email/templates      | Email templates      | Administrator | Database-backed override editor for the 10 transactional emails |
| Access control and system | /admin/organization-profile | Organisation profile | Administrator | Edition and organisation profile, with about content            |
| Access control and system | /admin/contact-inquiries    | Contact inquiries    | Administrator | Inbox for the public contact-us form                            |
| Access control and system | /admin/operations           | Operations           | Administrator | Operational toggles, for example the registration gate          |
| Access control and system | /admin/operation-log        | Operation log        | Administrator | Security-relevant business-event log                            |
| Access control and system | /admin/logs                 | Logs                 | Administrator | System logs                                                     |
| Access control and system | /admin/statistics           | Statistics           | Administrator | Event statistics                                                |
| Access control and system | /admin/attendance           | Attendance           | Administrator | Attendance records                                              |

The table below accounts for all 113 Control Panel routes. The sign-in flow, account pages, deep-link create fallbacks and framework error pages are absent from the navigation, reached by redirect or from the account menu.

| Route set                    | Routes                                                                                                                                                                                          | Count |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Main navigation pages        | Catalogued above                                                                                                                                                                                | 78    |
| Module placeholder           | /m/{module}, served by ModulePlaceholder                                                                                                                                                        | 1     |
| Reporting pages              | Nine pages under /admin/reports                                                                                                                                                                 | 9     |
| Further administrative pages | /admin/walk-in-mode, /admin/editions, /admin/visitors/badge-batches, /admin/sessions/live-hall, /admin/announcements, /admin/site-settings, /admin/ops/services, /admin/delegation-availability | 8     |
| Sign-in flow                 | /login, /login/totp, /login/recovery, /login/enrol-2fa, /forgot-password, /reset-password, /auth/pending, /auth/rejected                                                                        | 8     |
| Account pages                | /account/profile, /account/notifications, /account/totp-pairing                                                                                                                                 | 3     |
| Deep-link create fallbacks   | /admin/admins/new, /admin/others/new, /admin/visitors/new                                                                                                                                       | 3     |



| Route set             | Routes                             | Count |
|-----------------------|------------------------------------|-------|
| Framework error pages | /Error, /not-found, /not-permitted | 3     |
| Total                 | All Control Panel routes           | 113   |

## 4.2 Website screens

The public Website is Blazor SSR, serving static informational content about the forum and its programme, which keeps it fast. All 19 routes are public. It has no sign-in, no registration and no account page, and stores no personal data: registration and visitor data live in the mobile application and the Control Panel. With no authentication surface it holds no session, no identity cookie and no per-page permission. Public reads go through SimfPublicClient.

Two interactive exceptions exist by design: the visitor assistant, which passes question text to the AI service without storing the visitor's identity, and /meeting/confirm, whose only credential is an opaque per-speaker token in the link. The top navigation carries five entries: About, Programmes, Speakers, Discover Saudi Arabia and Archive, off-canvas on small screens. About and Programmes open drop-downs. Discover Saudi Arabia links out to [visitsaudi.com](https://visitsaudi.com). The pages share the bilingual Arabic and English right-to-left design system and the In- SSR component kit.

| Route                   | Screen name                  | Audience             | Purpose                                                                                            |
|-------------------------|------------------------------|----------------------|----------------------------------------------------------------------------------------------------|
| / and /landing          | Landing                      | Public               | Marketing landing page                                                                             |
| /about                  | About the forum              | Public               | The forum and the platform it provides                                                             |
| /about/objectives       | Objectives                   | Public               | Strategic objectives                                                                               |
| /about/organizer        | The organizer                | Public               | Organising body and the platform it convenes                                                       |
| /about/themes           | Key themes                   | Public               | Topics and issues across the sessions and workshops                                                |
| /about/venue            | The venue                    | Public               | Forum venue                                                                                        |
| /programme              | Programme                    | Public               | Published programme                                                                                |
| /programme/opening      | Opening ceremony             | Public               | Opening ceremony                                                                                   |
| /programme/sessions     | Plenary sessions             | Public               | Plenary session listing                                                                            |
| /programme/exhibition   | The exhibition               | Public               | Exhibition and its booths                                                                          |
| /programme/gov-meetings | Government business meetings | Public               | Bilateral meetings alongside the forum                                                             |
| /sessions/{id}          | Session detail               | Public               | One session, its speakers and downloads                                                            |
| /speakers               | Speakers                     | Public               | Speaker listing, on live data                                                                      |
| /partners               | Partners and sponsors        | Public               | Sponsors and partners by tier                                                                      |
| /archive                | Forum archive                | Public               | Past editions, milestones and material                                                             |
| /discover               | Discover Saudi Arabia        | Public               | Kingdom destinations, with an outbound link to <a href="https://visitsaudi.com">visitsaudi.com</a> |
| /visit                  | Visiting and travel          | Public               | Planning a visit and travel to the Kingdom                                                         |
| /privacy                | Privacy policy               | Public               | Personal-data notice, including how an account is deleted                                          |
| /meeting/confirm        | Meeting confirm              | Public, opaque token | Token-based meeting confirmation, the site's only credentialed entry point                         |

## 4.3 Mobile application screens

The mobile application is the Flutter project at `src/Mobile/simf_app`, built on Riverpod, `go_router` and Dio. A persistent five-tab bottom-navigation shell of Home, Sessions, Badge, Map and Profile sits in one

IndexedStack under a single route, so per-tab state is preserved. The auth and role gates for staff and moderator redirect the user.

The theme is navy always, with dark pinned. The application is bilingual and switches to right-to-left automatically for Arabic. Every in-app page carries the same app-bar invariant: notifications bell, language toggle, dark-mode indicator and hamburger. There are 42 feature folders under lib/features/. The catalogue below lists every shipped screen, grouped by feature area, a superset of those folders.

| Area                            | Screen (route)            | Audience             | Purpose                                          |
|---------------------------------|---------------------------|----------------------|--------------------------------------------------|
| Onboarding and entry            | splash (#1)               | Guest                | Launch and the server version-policy update gate |
| Onboarding and entry            | onboarding (#2)           | Guest                | Three-step first-run carousel                    |
| Onboarding and entry            | guestMode (#12)           | Guest                | Guest-mode landing                               |
| Authentication and registration | signIn (#3)               | Guest                | Sign-in                                          |
| Authentication and registration | verifyOtp (#3a)           | Mid-sign-in          | Sign-in second factor, an e-mailed OTP           |
| Authentication and registration | forgotPassword (#3b)      | Guest                | Request a reset OTP                              |
| Authentication and registration | resetPassword (#3c)       | After forgot         | OTP and new password                             |
| Authentication and registration | signUpForm (#5)           | Guest                | Account sign-up form                             |
| Authentication and registration | emailOtp (#6)             | Guest                | Sign-up e-mail verification                      |
| Authentication and registration | signUpVisitor (#7)        | Visitor              | Profile data, type and lookups                   |
| Authentication and registration | signUpInterests (#7-01)   | Visitor              | Interests and the single profile save            |
| Authentication and registration | registrationSuccess (#10) | Visitor (pending)    | Terminal sign-up confirmation                    |
| Authentication and registration | registrationStatus (#11)  | Visitor (pending)    | Account approval status                          |
| Authentication and registration | identityVerification      | Visitor (approved)   | Avatar liveness capture                          |
| Authentication and registration | biometricStepUp (#7a)     | Visitor+ (approved)  | Enrol a biometric through an emailed OTP         |
| Authentication and registration | badgeSignIn               | Guest (badge holder) | Badge-QR sign-in entry                           |
| Authentication and registration | badgePassword             | Guest (badge holder) | Badge-QR sign-in password step                   |
| Authentication and registration | badgeActivation           | Guest (badge holder) | Passwordless badge activation                    |
| Home, profile and information   | home (#13)                | Guest+               | Home dashboard                                   |
| Home, profile and information   | myArea (#14)              | Visitor              | My-area dashboard, with .ics and .vcf export     |
| Home, profile and information   | more (#41)                | Guest+               | Profile card, grouped navigation and sign-out    |
| Home, profile and information   | badge (#32)               | Signed-in            | QR entry badge                                   |
| Home, profile and information   | notifications (#33)       | Signed-in            | Notification list                                |
| Home, profile                   | accessibility (#38)       | Guest+               | Accessibility settings, applied app-wide         |

| Area                          | Screen (route)              | Audience        | Purpose                                                     |
|-------------------------------|-----------------------------|-----------------|-------------------------------------------------------------|
| and information               |                             |                 |                                                             |
| Home, profile and information | aboutForum (#37)            | Guest+          | The forum, its mission, vision and themes                   |
| Home, profile and information | aboutApp (#207)             | Guest+          | The app, the manual update check and support                |
| Home, profile and information | forumGuide (#200)           | Guest+          | Static forum guide                                          |
| Home, profile and information | faq (#201)                  | Guest+          | Public FAQ accordion                                        |
| Home, profile and information | contactUs (#203)            | Guest+          | Contact form and organisation-profile panel                 |
| Home, profile and information | terms (#9)                  | Guest           | Terms content                                               |
| Home, profile and information | chatbot (#36)               | Guest+          | AI assistant, grounded on the programme, FAQ and booth list |
| Programme and sessions        | sessions (#16)              | Guest+          | Agenda, with the day strip and type tabs                    |
| Programme and sessions        | sessionDetail (#17)         | Guest+          | Session detail, with the seat card                          |
| Programme and sessions        | mySeat (#18)                | Visitor         | My reserved seat                                            |
| Programme and sessions        | seatPicker                  | Visitor         | Seat selection or a random reserve                          |
| Programme and sessions        | joinSessionHub              | Visitor         | Join-session hub                                            |
| Programme and sessions        | liveBroadcast (#25)         | Login-only      | Live broadcast, carried on YouTube                          |
| Programme and sessions        | aiSummary (#34)             | Guest+          | AI session summary                                          |
| Programme and sessions        | sessionSummaryList (#111)   | Guest+          | Searchable session-summary list                             |
| Programme and sessions        | sessionPresentations (#202) | Approved        | Day-tabbed session and presentation list                    |
| Programme and sessions        | myAreaSessions (#113)       | Approved        | My-sessions, across four tabs                               |
| Programme and sessions        | sendQuestion (#26)          | Visitor         | Submit an audience question                                 |
| Programme and sessions        | rate (#40)                  | Visitor         | Dynamic feedback and rating form                            |
| Speakers                      | speakers (#19)              | Guest+          | Speaker list                                                |
| Speakers                      | speakerProfile (#20)        | Guest+          | Speaker profile, with the meeting request                   |
| Exhibition and delegations    | venueMap (#15)              | Guest           | 2D venue map and its booths                                 |
| Exhibition and delegations    | booths (#22)                | Guest+          | Booth list                                                  |
| Exhibition and delegations    | boothMap                    | Guest+          | One booth on the venue map                                  |
| Exhibition and delegations    | exhibitorDetail (#220)      | Guest+          | Exhibitor detail                                            |
| Exhibition and delegations    | sponsors (#23)              | Guest+          | Sponsor list                                                |
| Exhibition and delegations    | sponsorDetail (#221)        | Guest+          | Sponsor detail                                              |
| Exhibition and delegations    | delegations (#21)           | Guest+ (public) | Invited-country delegations                                 |
| Media, news                   | news (#29)                  | Guest+          | News hub, with the latest-updates tab                       |

| Area                              | Screen (route)         | Audience                                 | Purpose                                                                                          |
|-----------------------------------|------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------|
| and archive                       |                        |                                          |                                                                                                  |
| Media, news and archive           | gallery (#30)          | Guest+                                   | Photo and video gallery                                                                          |
| Media, news and archive           | mediaPartners (#31)    | Guest+                                   | Media-partner grid                                                                               |
| Media, news and archive           | archive (#24)          | Guest+                                   | Past-edition archive                                                                             |
| Media, news and archive           | archiveDetail (#24-01) | Public                                   | Archive edition detail                                                                           |
| Networking, contacts and meetings | meetPeople (#35)       | Visitor                                  | Smart match suggestions                                                                          |
| Networking, contacts and meetings | shareMyContact         | Visitor (approved)                       | Share a contact QR or vCard                                                                      |
| Networking, contacts and meetings | scanContact            | Visitor (approved)                       | Scan and save a contact                                                                          |
| Networking, contacts and meetings | myContacts             | Visitor (approved)                       | Saved contacts                                                                                   |
| Networking, contacts and meetings | meetings (#116)        | Entitled (approved; either meeting flag) | Bilateral meetings, gated by the per-user AllowsSpeakerMeeting and AllowsDelegationMeeting flags |
| Networking, contacts and meetings | requests (طلباتي)      | Visitor (approved)                       | Unified feed of document, badge and meeting requests                                             |
| Staff, moderator and exhibitor    | sessionModerate        | Moderator+ (per-session grant)           | Moderator Q&A console                                                                            |
| Staff, moderator and exhibitor    | gateScanner            | Staff (GateOperator grant)               | Staff gate scan console                                                                          |
| Staff, moderator and exhibitor    | staffRegisterVisitor   | Staff (Visitors.RegisterOnsite)          | Walk-in visitor registration                                                                     |
| Staff, moderator and exhibitor    | myVisitors (زوار)      | Exhibitor (approved)                     | Captured visitors list                                                                           |
| Staff, moderator and exhibitor    | scanVisitor            | Exhibitor (approved)                     | Exhibitor lead-capture scan                                                                      |

Six further screens ship outside the catalogue above, listed here so the inventory is complete.

| Screen (route)        | Purpose                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------|
| myInterests           | My-Area interests editor                                                                   |
| myMobile              | My-Area mobile-number editor                                                               |
| myDevices             | Enrolled biometric device keys, at /account/my-devices                                     |
| newsArticle           | Single news article, opened from the news list                                             |
| meetingConfirm (#117) | Other-party meeting confirmation, opened from a notification deep link at /meeting-confirm |
| staffSeating (#118)   | Staff guest-seating desk, at /staff/seating/{sessionId}                                    |

Four screen-sequence numbers are unused and not part of the shipped application: signUpType (#4), savedSessions (#8), audienceComments (#28) and myMeetings (#115). The Flutter screens catalogued here are the shipped user interface over the live /app/\* API.

## 5. Module Description

This section specifies each SIMF module: its purpose, functions, business rules, interfaces and data, and the errors it returns. The rules stated here are the rules the build enforces.

The backend organises the modules as feature folders under `src/Backend/SIMF.Api/Endpoints`, with matching infrastructure services under `src/Backend/SIMF.Infrastructure`. The application surface is served under `/api/v1/app`, the administrative surface under `/api/v1/admin`. Each description states whether its module is presented on the mobile application, the Control Panel, the website or a combination.

Cross-cutting behaviour is stated once here rather than fourteen times. Every response carries the `ApiResponse<T>` envelope. Error messages are bilingual, Arabic and English. A `FluentValidation` failure returns `VALIDATION_FAILED` with HTTP 400 and one detail per field. Administrative endpoints and Control Panel pages are permission-gated. Configuration and business events reach the operation log.

### 5.1 Authentication & Login

**Purpose.** The module owns sign-up, sign-in and session lifetime: it carries a new account to `EmailVerified` and issues a session to an approved user. SIMF runs its own identity store: no federation with a ministry directory, no single sign-on, no Active Directory, no LDAP and no NAFATH, the users being members of the public rather than ministry personnel.

Key functionalities:

- Sign-up, creating a Registered account and e-mailing a six-digit code.
- E-mail verification and code resend.
- Password sign-in.
- A second factor: TOTP where an authenticator key is paired, an e-mailed OTP otherwise, with ten single-use Crockford recovery codes as the TOTP fallback.
- Token refresh with rotation, and sign-out.
- Forgotten-password reset and self-service password change.
- TOTP enrolment on the two-slot pattern, and administrator reset of a user's second factor.

Business rules:

- Password policy: at least 8 characters, at least 1 letter and 1 digit, not the e-mail address.
- Sign-in branches on account state: `Registered` refused with `AUTH_EMAIL_NOT_VERIFIED`; `EmailVerified` and `Approved` allowed; `PendingApproval` allowed with limited access on the website and the application, refused on the Control Panel; `Rejected` and `Disabled` refused.
- Invalid-credential and forgotten-password responses are generic and never disclose whether an account exists.
- The access token lives 5 minutes. The refresh token defines an absolute 24-hour session that rotation does not slide.
- A sign-in audience gate over `cp`, `web` and `app` admits only Control Panel role holders to the Control Panel, only visitors to the website and the application.
- Account lockout and per-code attempt caps bound brute-force attempts.

Interfaces and data. The module exposes the /app/auth endpoints sign-up, verify-email, resend-code, sign-in, verify-totp, verify-otp, verify-recovery-code, refresh, sign-out, forgot-password and reset-password, with /app/auth/change-password and the administrative POST /admin/admins/reset-two-factor. It uses IJwtTokenService with its JwtTokenService implementation, the Identity store, the OTP and TOTP verifiers, and the e-mail channel of Notifications for verification and reset codes. Registration and Approval begins where it ends, at EmailVerified, and every client depends on it for session tokens.

Error handling. The module returns bilingual ApiError codes:

- AUTH\_EMAIL\_ALREADY\_REGISTERED, HTTP 409.
- AUTH\_CODE\_INVALID and AUTH\_CODE\_EXPIRED, HTTP 400.
- AUTH\_ACCOUNT\_NOT\_FOUND, HTTP 404.
- AUTH\_INVALID\_CREDENTIALS, AUTH\_EMAIL\_NOT\_VERIFIED, AUTH\_ACCOUNT\_NOT\_APPROVED, AUTH\_ACCOUNT\_DISABLED and AUTH\_ACCOUNT\_LOCKED.
- AUTH\_MFA\_TOKEN\_INVALID and AUTH\_MFA\_TOKEN\_EXPIRED, and AUTH\_TOTP\_INVALID.
- AUTH\_REFRESH\_TOKEN\_INVALID and AUTH\_REFRESH\_TOKEN\_EXPIRED.
- AUTH\_RESET\_CODE\_INVALID and AUTH\_RESET\_CODE\_EXPIRED.
- AUTH\_WRONG\_SURFACE\_CP and AUTH\_WRONG\_SURFACE\_WEB.
- RATE\_LIMIT\_EXCEEDED, HTTP 429.
- A rotated refresh token presented twice is rejected and logged as a possible theft signal.

## 5.2 Registration & Approval

Purpose. The module carries an EmailVerified account through registration, the organisers' review, the assignment of a final user type and the issue of an entry badge, for the Visitor, Other and Exhibitor registration types.

Key functionalities:

- Choice of registration type.
- Capture of personal, identity and contact details and attachments.
- Identity photo verification, with the alternative identity-photo path for women.
- The exhibitor organisation branch.
- Terms consent, submission and registration-status tracking.
- Security review of Visitor and Other registrations, and Public Relations approval of exhibitors with booth assignment.
- Assignment of the final user type and issue of the badge.
- On-site registration and badge reprint.
- Opening and closing registration.
- Internal-user creation, with TOTP enrolment at first sign-in.
- Administrator creation of a user with a 7-day invite code.
- Separate Control Panel pages for staff and visitors.
- The PendingApproval to Approved or Rejected workflow, with a rejection reason of 10 to 500 characters.
- The 12-character Crockford QrId minted on approval.
- The identity image encrypted at rest under AES-256-GCM.

- Bulk administrative actions.

Business rules:

- Submission requires every mandatory field and terms consent. The form branches by registration type and adapts the identity fields for Saudi and non-Saudi applicants.
- A Saudi national identity number is 10 digits beginning with 1. An Iqama number is 10 digits beginning with 2. A passport number is free-form.
- A Saudi mobile number matches 05XXXXXXX or +9665XXXXXXX, and an international mobile number matches E.164.
- A mobile number is stored canonically rather than as typed: separators stripped, a leading 00 rewritten to +, and the Saudi local spelling folded onto +9665XXXXXXX.
- The identity image is a required attachment.
- QrId is minted at the PendingApproval to Approved transition, not at record creation.
- Rejection requires a recorded reason.
- Registration closes automatically when the administrator-set auto-close time on the RegistrationGate singleton passes; a closed state blocks submission.
- The final user type set at approval drives permissions and application access, and ProfileType.MobileAppRole resolves the mobile\_app\_role claim in the JWT.

Interfaces and data. The module exposes `/app/account/user-profile` with its countries and identity-image sub-resources, `/app/account/profile-types`, and the administrative `/admin/admins` and `/admin/visitors` families: list, create, `{id}/approve`, `{id}/reject`, `pending/list`, and bulk delete, duplicate, import and export. It issues the badge `Badge` and Access Control operates, and raises registration-submitted, registration-approved, registration-rejected and badge-ready events to Notifications. The encrypted identity image is stored through `EncryptedUserIdDocumentStorage`, avatars through `IAvatarStorage`.

Error handling:

- `AUTH_ACCOUNT_NOT_FOUND` for an unknown account.
- `ADMIN_EMAIL_ALREADY_REGISTERED` for an e-mail already held.
- `ADMIN_USER_NOT_PENDING`, HTTP 409, on a repeated approval or rejection.
- `VISITOR_NATIONALITY_UNKNOWN` for an unrecognised nationality.
- `VALIDATION_FAILED` per field.
- Uploads are magic-byte sniffed and capped at 5 MB for the identity image and 2 MB for the avatar, and rejected before persistence.
- A bulk operation skips an invalid target row by row, such as a self-deletion or an administrator acting on another administrator, rather than failing the batch.

### 5.3 Badge & Access Control / Gates

Purpose. The module operates the badge issued at approval: it shows the badge to the attendee, verifies it at the venue, exchanges attendee contacts and records hall arrival.

Key functionalities:

- The badge card and its signed-token QR.

- Venue-entry verification by a Staff scan.
- Attendee-to-attendee contact exchange by badge scan.
- Hall-arrival capture by a QR scan at the hall door.
- The Gate Module: a configurable access point with a 13-step ordered constraint engine, allow-lists, idempotency, a duplicate-absorption window, a derivation of who is currently inside, and a failure-rate circuit breaker.

Business rules:

- The badge QR is a signed token verified on the server. A copied or invented QR fails.
- A scan is allowed only for an active badge whose holder is Approved.
- Hall arrival is recorded by a QR scan at the hall door. Automatic arrival from a GPS geofence is deferred, open item G-OI-2.
- Where both means record an arrival for one attendee and session, they update a single HallAttendance row; one open attendance row per session is enforced.
- The gate engine runs its steps in a load-bearing order and short-circuits to a recorded denial carrying a named DenialReasonCode.
- An empty allow-list passes. An allow-list that filters to empty denies everything, which is the safe default.
- A 5-second window absorbs a double fire of the same scan.
- 10 or more backend faults in 60 seconds open a 5-minute circuit. A recorded policy denial never counts towards it.
- In the deferred geofence chain, GPS presence is reported every 30 to 60 seconds in small batches, never as a continuous stream.
- GateScan is append-only.

Interfaces and data. The module exposes POST /app/gates/{gateId}/scans, GET /app/gates/my-reports/today and the administrative /admin/gates endpoints. It consumes IqResolver and the idempotency store ScanIdempotency, and reads User, UserProfile, ProfileType, Hall and Session. It writes HallAttendance, VenueEntry and GateScan, read by Statistics and by the question gating in Engagement, and records a SavedContact for Contacts and Sharing.

Error handling. Denials are recorded against the scan:

- Recorded denials: QR\_UNKNOWN, GATE\_INACTIVE\_AT\_SCAN, HOLDER\_NOT\_APPROVED, HOLDER\_DISABLED, HOLDER\_LOCKED, PROFILE\_TYPE\_INACTIVE and PROFILE\_TYPE\_NOT\_ALLOWED.
- Two further denial codes are defined and reserved: OUTSIDE\_TIME\_WINDOW and BOOKING\_REQUIRED\_MISSING.
- Transport failures are not recorded as denials: HTTP 401 for an unauthenticated call, GATE\_OPERATOR\_NOT\_ASSIGNED with HTTP 403, IDEMPOTENCY\_KEY\_CONFLICT with HTTP 409, and GATE\_FAILURE\_CIRCUIT\_OPEN with HTTP 429.
- Every denial also emits one OperationLog row. A successful scan is recorded in GateScan only.

## 5.4 Forum Programme & Sessions

Purpose. The module defines the event content: the themes and pillars with their sub-topics, the halls and their seating, the speakers and their presentations, and the sessions tying a theme, a hall and speakers to



a time. It presents the attendee agenda and is the content bookings, live, engagement, AI-summary and statistics build on.

Key functionalities:

- Control Panel management of themes and sub-topics.
- Halls, each with a generated seat grid.
- Speaker profiles with presentation files.
- Sessions: bilingual title and description, theme, hall, category, start and end time, one or more speakers each with a role, and a live or non-live flag.
- The attendee agenda: day selector, search, session detail, speaker profile, add to calendar and device reminders.

Business rules:

- All programme content is required in both Arabic and English.
- Saving a hall capacity generates one seat per position. Reducing it removes only unallocated seats and warns about seats already assigned.
- A geofence per hall belongs to the deferred arrival chain, open item G-OI-2.
- A session's times fall within the forum dates, end after they start, and do not overlap another session in the same hall.
- A session carries at least one speaker with a role.
- A session marked live is streamed by the Engagement module.
- Programme records are soft-deleted. A theme still referenced is not removed.

Interfaces and data. The module exposes GET /app/programme/days and GET /app/programme/sessions/{id}, and the administrative POST, PUT and DELETE /admin/sessions with the equivalent theme, hall, speaker and presentation endpoints. It reads Category for the session category, stores speaker photographs and presentations through ISpeakerPresentationStorage, provides the sessions, halls and seats consumed by Bookings, the Exhibition venue map, Engagement and Statistics, and raises session-reminder events to Notifications.

Error handling. VALIDATION\_FAILED covers a missing bilingual field, an end time before the start time, a session outside the forum dates, and a hall already booked for the time window, which is a hall clash. Programme record changes are logged.

## 5.5 Bookings & Attendance

Purpose. The module lets an approved attendee reserve a session seat and reports whether the attendee attended. A reservation carries no payment: attendance is free, and the system has no payment, ticketing, invoicing or finance capability anywhere.

Key functionalities:

- Making a booking from the seat map.
- One attendee's bookings may not overlap in time.
- Cancellation before the session starts.
- The seat map and the My Seat view.

- Session attendance derived from HallAttendance.
- Seat allocation by user booking, random assignment, administrator-reserved rows and open seating.

Business rules:

- The attendee is Approved, the session is open with a free seat, and the session does not overlap a Pending or Approved booking of the same attendee.
- A booking is created Approved: the reservation auto-confirms and the seat is held. Gate check-in confirms it; a sweep before the session start releases any hold that has not checked in.
- Cancellation is allowed only before the session start time.
- A seat cannot be held or confirmed twice for one session. A filtered unique index enforces this, and a constraint-violation handler reports the seat as just taken.
- Booking.Status includes a Rejected value. No administrative endpoint for approving or rejecting a booking is mapped.
- The BookingConfirmed and BookingRejected notification kinds are defined. An attendee reservation auto-confirms and raises no booking notification.
- Live counts are computed by aggregating attendance on a short cycle, not by incrementing a hot counter row.

Interfaces and data. The module reads the sessions, halls and seats provided by Forum Programme and the HallAttendance rows written by Badge and Access Control. It raises the session-started-without-attendance and session-started-without-venue-entry events to Notifications.

Error handling:

- An overlapping booking is blocked with an explanation.
- A seat taken in the meantime asks the attendee to choose again.
- A full session reports that no seats remain.
- Cancellation after the session has started is refused.
- A rejection recorded without a reason returns VALIDATION\_FAILED.
- A uniqueness-constraint violation under concurrency is surfaced as a clear message rather than a 500.

## 5.6 Exhibition

Purpose. The module presents the exhibition: the booth directory, the sponsors and their tiers, and the interactive 2D venue map, so an attendee finds an exhibitor and navigates the venue.

Key functionalities:

- Control Panel management of booth records: hall, number, logo, bilingual descriptor and booth contact.
- Control Panel management of sponsors: bilingual name and description, logo and tier.
- Control Panel management of venue-map nodes. The map is 2D, each node a VenueMapNode with a Kind and an X and Y position.
- A searchable booth directory with a directions action and dialler and mail links.
- Sponsors grouped by tier.

- A pan and zoom map with booth preview popups and in-venue navigation to a booth or an assigned seat.

Business rules:

- A booth record exists once an exhibitor is approved and a booth assigned. Registration and Approval owns that step; this module manages and presents the record.
- A booth number is unique within its hall.
- A sponsor tier is one of the configured tiers. The SponsorTier values are Platinum, Gold, Silver and Bronze.
- All content is bilingual. Booth, sponsor and map content is public to read and requires authorisation to change.

Interfaces and data. The module exposes GET /app/booths and GET /app/sponsors, and the administrative POST /admin/exhibitors/list and POST /admin/sponsors. It reads ExhibitorProfile, Hall and Seat and the logo assets. The map guides an attendee to the seat assigned by Bookings and links a hall to its sessions in Forum Programme. Attendee position falls under the location-privacy rules.

Error handling. A duplicate booth number is rejected, and a missing bilingual field returns VALIDATION\_FAILED. Booth, sponsor and map-node changes are logged.

## 5.7 Engagement, Live, Q&A, Ratings

Purpose. The module lets the audience take part in a session: the live broadcast with an optional notification to the session's seat-holders, the questions an attendee puts to a moderator, and the multi-trigger ratings of a session, a day, the programme, the exhibition and the application.

Key functionalities:

- Starting and stopping the live broadcast. The stream is carried on YouTube through youtube\_player\_iframe, with HLS or MP4 as the fallback.
- A caption-language picker on the player.
- An optional notification to the session's seat-holders when a broadcast starts.
- A three-stage question pipeline across the Pre and Live phases: an advisory AI filter, then the Scientific Committee queue in the Control Panel, then the per-session moderator desk.
- Rating prompts at end of day, end of programme, end of session by the clock, on leaving a session view, on hall departure and on closing the live stream.

Business rules:

- Questions close at session end and have no lower bound, so a question may be asked before the session starts.
- Once live, an arrival gate applies only at a hall carrying a geofence: the asker must hold a HallAttendance record for the session. A hall without a geofence accepts the question, presence being unverifiable. The geofence is deferred, open item G-OI-2.
- A question moves from Pending to Approved or Hidden, an approved one pushed on stage through IsPushed. A moderator acts only on assigned sessions.

- The AI question filter is advisory only, gated by the configuration key `SessionQuestions:AIFilterEnabled`, whose default is the stub.
- A broadcast can notify the session's seat-holders or a chosen audience. There is no location-based targeting, and SIMF applies no geographic restriction to viewing the live stream.
- A rating fires once per session per user and is de-duplicated across the triggers.
- Sign language and live translation are withdrawn and not delivered. Two endpoints remain in the API, no client calls them, and their removal is an open item.

Interfaces and data. The module exposes the application session-question, moderator-desk and `/rate` endpoints, and the Scientific Committee queue at `/admin/questions/queue`. It reads Session, User and HallAttendance, and routes the AI filter through the shared `IAIService` cognitive-AI abstraction. Real-time updates are delivered over REST and polled on a bounded interval. It raises live-session-starting and rating-request notifications to Notifications.

Error handling:

- The question composer is hidden after the session ends, by design rather than as an error.
- A live-phase question at a hall carrying a geofence is refused with `NOT_AT_VENUE`, HTTP 403, when the asker holds no arrival record.
- A moderator opening the queue of an unassigned session is refused.
- When the AI filter is unavailable the result falls back to `ai-unavailable` and the question still reaches the Scientific Committee queue.

## 5.8 Networking and Cognitive AI

Purpose. The module connects attendees with one another and provides the cognitive-AI capabilities behind one AI-provider abstraction: subtitle import from the published session video, with manual paste or upload as fallback; the AI session summary that becomes the session minutes, drafted automatically then reviewed, approved and published by the Scientific Committee before attendees see it; question filtering before the moderator queue; the FAQ-backed assistant in the application and on the website; interests and matchmaking; and one-to-one meeting requests. Every AI feature runs on the GPT OSS 120B model, hosted by SITE on the on-site LLM server in HSA.

Key functionalities:

- Interest selection from the dynamic Category lookup.
- Matchmaking that suggests people with shared interests and sessions, carrying a match score and reason, and raising a recommendation and a push notification.
- One-to-one meeting requests routed to Public Relations for approval.
- A conversational AI assistant grounded on the whole programme, FAQ and booth list.
- AI session summaries stored as `SessionSummary` and produced through the central `IAIService`.
- The networking `Connection` entity and the speaker and delegation `MeetingRequest` entity.
- A Control Panel AI-settings page.

Business rules:

- A meeting request carries a topic and names an existing Approved target other than the requester.

- A decline records a reason. The request moves from Pending to Approved or Declined, appearing on approval in both attendees' areas.
- Matchmaking exposes only what the other attendee's profile already shows. A recommendation triggers at a score greater than or equal to 80%.
- The AI assistant answers free text from the event content it is given, its system prompt restricting it to that content.
- FAQ groups and entries are bilingual and managed in the Control Panel; the FAQ screen is where an attendee browses them.
- Every AI capability runs through the cognitive-AI abstraction. The provider is configuration rather than code, and only the data a capability needs is passed to it.
- The AI is never the final authority on a question. The Scientific Committee and the session moderator decide.

Interfaces and data. The module exposes POST /app/account/connections, POST /app/speakers/{id}/meeting-requests, the AI-assistant, FAQ and AI-summary endpoints, and the administrative POST /admin/business-meetings/list and POST /admin/delegation-meeting-requests/list with the FAQ and AI-settings pages. It reads User, Session and Category, shares the AI abstraction with the question filter and the captions in Engagement, and raises match-recommendation and meeting notifications to Notifications.

Error handling. A meeting request to an inactive, unknown or self target, or one without a topic, is rejected with a clear error, and a decline requires a reason. AI capabilities degrade behind the abstraction: provider and endpoint are configuration values, here the on-site GPT OSS 120B server reached over its OpenAI-compatible API. When it is unreachable the filter, subtitle and summary paths report a clear error rather than failing silently.

## 5.9 Notifications

Purpose. The module is the single notification abstraction the rest of the system reaches users through. It offers one operation, notify this recipient about this event type with this content, over four channels: in-app, e-mail, SMS and WhatsApp. It owns the in-app inbox and the reminders.

Key functionalities:

- A calling feature raises an event. The service builds a Notification, sends it on the channels configured for its type, and records one NotificationDelivery per channel.
- The in-app inbox with an unread count, filters, mark as read and a deep link.
- Session and attendance reminders. The session-reminder job de-duplicates on Session.ReminderSentAt.
- Per-type channel-mix configuration.
- Asynchronous sending. The API drains the e-mail queue in process through MailKit.

Business rules:

- A calling feature never chooses channels and never formats a channel message; the abstraction does both.
- The channel mix per type is configuration rather than code, and every type has at least one channel.

- A notification is sent only to its intended recipient and in the recipient's language. Verification and reset codes go by e-mail only.
- Sending is asynchronous, so a user request never blocks on a gateway round trip.
- A failed channel send is recorded and retried with bounded backoff behind a per-adapter circuit breaker.
- NotificationKind is an enumeration persisted by name and extended additively, for example BookingConfirmed = 40, SessionReminder = 41 and MeetingScheduled = 43.

Interfaces and data. The module exposes the in-app inbox reads and the administrative compose and channel-mix pages. It consumes events from Authentication, Registration, Bookings, Engagement and Networking, and sends through SmtplibEmailSender and EmailBackgroundService; the SMS and WhatsApp adapters are deferred and not part of this delivery. In-app push is read by the client from the inbox over REST.

Error handling. A failed channel send is captured as a NotificationDelivery with a failure status and retried. Adapter timeouts and a circuit breaker bound a slow or unavailable gateway. Validation requires a type from the catalogue, an existing recipient, at least one configured channel, and a title and body in the recipient's language.

## 5.10 Media, News & Archive

Purpose. The module holds the content the forum publishes about itself and its history: the Media Center with its media items, gallery and media partners, the news items with their categories, and the archive of previous editions with its post-event visibility control.

Key functionalities:

- Control Panel management of media items: kind (post, photo, video or social post), bilingual title and body, asset and publish date.
- Control Panel management of media partners: name and logo.
- Control Panel management of news items: bilingual title and body, image, publish date and category.
- Control Panel management of previous editions: title, brief, sessions, place, time, image, video, past speakers and edition statistics, each carrying an IsVisible flag.
- Read-only presentation on the website and the application: gallery, social content, media-partner directory, news stream and one archive card per year.

Business rules:

- All content is bilingual and required in both languages.
- A news item carries an active news category and an image.
- A media item's asset matches its kind.
- An edition year is unique.
- The current edition, SIMF 2026, is hidden from the archive until the event ends, under its visibility flag. Past editions are visible.
- Social content is reviewed before it is published.
- Content is public to read and requires authorisation to change.
- The media and news field sets and the news category list are open: the client confirms them.

Interfaces and data. The module exposes GET /app/news and GET /app/media, and the administrative POST /admin/news and POST /admin/media/list with the media-partner and previous-edition endpoints. It reads Category for the news categories and stores images, videos and logos through the media and asset storage services. The archive owns ArchiveMediaItem, ArchiveSessionTitle and ArchivePastSpeaker, and the ArchiveVisibility singleton.

Error handling. A missing bilingual field, a missing category or image, or a duplicate edition year produces VALIDATION\_FAILED. Content and visibility changes are logged.

### 5.11 Statistics & Dashboards

Purpose. The module presents the figures organisers watch before and during the forum, and the live picture of who is in the venue, on the Control Panel dashboard.

Key functionalities:

- The dashboard page.
- Per-day figures: registrations, badges printed, VIP and media badges, check-ins and approximate entries.
- Overall figures: themes and topics, speakers, participating countries, registrations and badges, student badges, check-ins per day and total attendance, broadcast hours, and audience questions.
- Live venue attendance and per-hall attendance.
- The GPS-presence movement and dwell-time views are deferred, open item G-OI-2.

Business rules:

- The Statistics context owns no source-of-truth records. It reads from the other contexts through their application services and never by a cross-context query.
- Figures are derived and never hand-entered. The one exception is student badges printed outside the Control Panel, open for confirmation.
- Every figure is computed on request by aggregating VenueEntry and HallAttendance through the owning contexts. Nothing is persisted: no statistics snapshot, no hot counter row.
- The dashboard reads GpsPresence rollups and never raw rows.
- GPS presence is sensitive personal data, collected only with permission and encrypted at rest. Its retention period is an open item.

Interfaces and data. The module exposes GET /admin/statistics. Through the owning-context services it reads RegistrationRequest, Badge, VenueEntry, HallAttendance, Session, SessionQuestion, Theme, Speaker and Category, and uses GpsPresence, whose table is DevicePositionPings. The live panels refresh over REST on a bounded interval.

Error handling. A role without the Dashboard permission is refused the dashboard route. A stale live panel shows its last-updated time. A heavy figure is recomputed on every read. A device whose owner denied the location permission contributes no GPS-presence data.

### 5.12 Control Panel Configuration

Purpose. The module lets organisers change content, categories, labels, colours, venue tracks and key settings without a release, and records every change. It splits across a Content and Categories page for

the content team and a System Configuration page for the technical team, with the Operation Log alongside.

Key functionalities:

- Dynamic content blocks: headings, texts, the in-app welcome message, banners, images, section labels and page content, bilingual and read at runtime.
- Dynamic categories, labels and colours, with add, hide and delete, a display order and a colour per category.
- Venue-track management.
- Opening and closing registration.
- The system configuration settings.
- The Operation Log viewer.
- The configuration surfaces are `SystemSetting` as a key and value pair, `OrganizationProfile`, `ContentBlock` and `Banner`, and the `RegistrationGate` singleton.

Business rules:

- A change here takes effect across the website and the application at runtime, with no release.
- A content block holds a key and a value in both languages. A category entry holds a bilingual name, a colour, a display order and a Kind.
- A category in use is hidden, not hard-deleted.
- Registration closes automatically when the administrator-set auto-close time passes.
- This module manages content and per-category colours only, not the brand palette or the typeface fixed by the visual identity and the theme tokens.
- The operation log is append-only. No user, including an Administrator, edits or deletes an entry. It is enforced at the database as well as in the application: the runtime account may insert but not update or delete.
- Operation-log entries are also shipped to the ministry log collector.
- Roles and permissions are not managed here.

Interfaces and data. The module exposes the administrative `POST /admin/system-settings/list` and `POST /admin/operation-log/list`, the content, category, venue-track and registration-control pages, and `GET /app/site-settings` so clients read content at runtime. Every other module writes its events to the operation log this module owns and displays. Content-block images are stored as assets.

Error handling. A user without the required permission opening a configuration page is refused. A log entry cannot be edited. A category in use is hidden rather than removed, so existing references hold. An uploaded content-block image is restricted by type and size.

### 5.13 Business Meetings

Purpose. The module runs administrator-arranged business-to-business and business-to-consumer meetings, and the per-hall configuration and allocation they sit on. An operator configures a hall's purpose and layout, reserves space over a time slot and schedules a meeting between named parties at a reserved table. Confirmed meetings surface on each participant's mobile dashboard.

Key functionalities:



- Per-hall purpose, which is Booth, Session or Meeting, and per-hall layout.
- Allocation by whole hall, random by count, or row and column, over a from and to slot.
- Meeting-table definition with a capacity.
- Scheduling a confirmed meeting, business-to-business or business-to-consumer, with a group of three or more allowed.
- Conflict enforcement.
- Cancellation with notification of the participants.
- The Control Panel hall-configuration and Business Meetings pages.
- A separate attendee-initiated speaker and VIP request path, with a hall-availability gate and a speaker double opt-in by e-mail.

#### Business rules:

- An administrator-arranged meeting is created Confirmed. There is no Pending state, and this path carries no attendee request queue.
- Participants number at least 2, are distinct, and each is active and approved. A group fits within the table capacity.
- One active reservation per unit per overlapping slot, enforced by a filtered unique index and a race guard. No participant is double-booked across overlapping meetings.
- Random-by-count allocation stops at the requested count or at the hall capacity, whichever is smaller.
- A participant is either a Company, held as a foreign key inside the application database, or a visitor, held as a bare Guid with no cross-database foreign key, resolved on read.
- The attendee-initiated speaker path adds an AwaitingSpeaker state and an anonymous action-token endpoint. The token is single-use, valid for 72 hours, and backs the approve and reject links in the speaker's e-mail.

Interfaces and data. The module exposes the administrative `/admin/meeting-tables`, `/admin/business-meetings`, `/admin/speaker-availability` and `/admin/speaker-meeting-requests/{id}/respond`, and the application `POST /app/speakers/{id}/meeting-requests`, `GET /app/speakers/{id}/available-slots` and `POST /app/delegation-meeting-requests`. It reuses the seat-reservation mechanics of Bookings and resolves visitor names by a round trip to the Identity store. It raises the `MeetingScheduled`, `MeetingCancelled` and `MeetingRequestConfirmed` notifications, and feeds the meeting union on the `Page_014` dashboard. The speaker e-mail uses the MailKit path.

#### Error handling:

- An inactive or unknown participant, a table already taken for an overlapping slot, or a participant already in an overlapping meeting produces a clear error.
- A group larger than the table capacity is rejected.
- An administrator without the permission receives HTTP 403 and does not see the navigation entry.
- A notification failure never rolls back the meeting: the failure is swallowed and logged.
- A used or expired action token shows a neutral landing page saying the link is no longer valid, and leaks no state.

## 5.14 Contacts & Sharing

Purpose. The module carries two deliberately separate mechanisms. Track 1 is a shared, de-duplicated Contact directory referenced by foreign key from Company, Sponsor, MediaPartner, Speaker and the Booth officer. Track 2 is consent-by-action visitor-to-visitor contact sharing, by QR or by an operating-system share-intent vCard, saved to My Contacts.

Key functionalities:

- Track 1: create, edit and soft-delete a Contact holding a logo, a bilingual name, phone numbers, a fixed social set, a website, map latitude and longitude, and a country foreign key.
- Track 1: link a Contact from any of the five organisation entities through a picker that links an existing record or creates one. One row is reused across roles, so editing it once updates every reference.
- Track 2: mint and rotate a dedicated share token and display it as a QR.
- Track 2: resolve a token to a card projected live, save a SavedContact, list and remove My Contacts, and export a vCard 3.0 through the operating-system share sheet.

Business rules:

- A visitor card and an organisation contact are never the same record. A visitor card is projected live from UserProfile, with a permitted bare Guid round trip to the Identity store for the e-mail address; copying it into the organisation directory is forbidden by the two-database rule.
- The share token is dedicated and separate from the entry QrId, so a gate scan never harvests a card, and is rotatable.
- A Contact is independent of the entities referring to it. Deactivating one an active entity still references is blocked by OnDelete.Restrict and a guard.
- An organisation read projection prefers the linked Contact and falls back to the inline columns, preserving the shipped append-only wire contract.
- Saving is idempotent per owner and subject. An attendee cannot save their own card. No personal-data snapshot is stored, the card being resolved on read, and the subject is not notified.
- The social links are a fixed set: Facebook, X, LinkedIn and Instagram. A logo is a relative path and never an absolute URL.

Interfaces and data. Track 1 exposes administrative CRUD and the picker, gated by Contacts.View and Contacts.Edit, wired into the five organisation admin forms, with one shared read-only card component on the website and the application. Track 2 exposes the application endpoints under /app: GET share-token, POST share-token/rotate, POST contacts/resolve, POST contacts/save, GET contacts, DELETE contacts/{id} and GET contacts/{id}/vcard. These are application-only with no Control Panel surface, gated by RequireApprovedAccount with the app audience, as Connection is. The module reads Country, UserProfile and Organisation, and does not replace the Connection entity of Networking.

Error handling:

- Soft-deleting a Contact that is still referenced is rejected.
- Resolving an unknown or revoked share token returns HTTP 404.
- An attempt to save one's own card is rejected.
- A deactivated subject makes the saved row show a limited or unavailable card.

- Field validation aligns the FluentValidation rules, the EF configuration and the user-interface lengths.
- Latitude and longitude are both set or both null, and within valid ranges.
- A social URL is validated as an absolute URL.

## 6. Data Model

This section gives the conceptual view by bounded context, the entity relationship diagram, and the schema design. The model is logical, with the sourced key columns for the dictionary tables.

Data classification. The platform holds no classified and no militarily sensitive information; its data is unclassified, in three groups: personal data of visitors, speakers and exhibitors; open scientific material, the sessions and workshops; and administrative data for exhibition and programme management. The personal-data group falls under the Saudi Personal Data Protection Law (PDPL) and the National Cybersecurity Authority controls, and carries the encryption-at-rest, access-control and audit obligations described here. No personal or operational data reaches any cloud artificial-intelligence service.

### 6.1 Conceptual View

The model is technology-agnostic, organised by the twelve bounded contexts. Each context owns its tables and reads another context's data through that context's application service, never through a cross-context query. Standard columns are stated once below and not repeated per entity.

Conventions.

- Every table has a GUID Id primary key.
- Tables and columns are PascalCase; table names are singular.
- Soft delete uses an IsActive boolean, which list queries filter on.
- Mutable entities carry CreatedAt, CreatedBy, ModifiedAt and ModifiedBy.
- A foreign key takes the form EntityId, naming the entity it points at.
- Fixed, software-owned sets are enums. Client-maintained sets, being registration sub-types, session categories and interests, are Category rows.
- Bilingual text uses paired columns, TitleAr and TitleEn, not a translations table.
- Timestamps are stored as Saudi local wall-clock time, with no conversion either way.

The twelve contexts, their entities and relationships:

| Context           | Entities                                                                                          | Relationships                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Identity & Access | User, Role, Permission, RolePermission, UserRole, RefreshToken, EmailVerificationCode, TotpSecret | User has many UserRole. Role has many RolePermission and UserRole. Permission is one action on one page from the fixed catalogue, with many RolePermission. RefreshToken, EmailVerificationCode and TotpSecret belong to User. The AccountState enum runs Registered, EmailVerified, PendingApproval, Approved, Rejected, Disabled. As built on ASP.NET Core Identity: User, Role and UserRole are SimfUser deriving from IdentityUser of Guid, SimfRole and UserRoles; TotpSecret is the Identity authenticator-key token store, with no separate table; EmailVerificationCode is AccountCode with |

| Context                 | Entities                                                                                                               | Relationships                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         |                                                                                                                        | a Purpose field.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Registration & Approval | RegistrationRequest, AttendeeProfile, ExhibitorProfile, Companion, Attachment                                          | RegistrationRequest belongs to User, with one final-type Category and many Attachment. AttendeeProfile and ExhibitorProfile belong to User, Companion to ExhibitorProfile. Attachment belongs to RegistrationRequest and references an Asset. The final user type is a Category of kind VisitorSubType or OtherType, set on approval.                                                                                                                                                                                  |
| Badge & Access Control  | Badge, VenueEntry, HallAttendance, SavedContact, Gate, GateProfileTypeAllow, GateAssignment, GateScan, ScanIdempotency | Badge belongs to User, VenueEntry to Badge. HallAttendance belongs to User and Session, recording enter and leave times by QR scan; the GPS-geofence method is deferred as open item G-OI-2. SavedContact links an owner User to a scanned User. Gate has many GateProfileTypeAllow, GateAssignment and GateScan. GateScan is an append-only audit log with a bigint identity primary key. Logical cross-context foreign keys to Users, UserProfiles and ProfileTypes are enforced at write time by the gate services. |
| Forum Programme         | Theme, SubTopic, Hall, Seat, Session, Speaker, SessionSpeaker, SpeakerPresentation, Booking                            | Theme has many SubTopic and Session; Hall has many Seat and Session. Session belongs to Theme, Hall and a session-category Category, with many SessionSpeaker. SessionSpeaker links Session and Speaker, carrying RoleInSession, either Speaker or Host. SpeakerPresentation belongs to Speaker and Session. Booking belongs to User, Session and Seat; a reservation is confirmed automatically on create and again at gate check-in.                                                                                 |
| Exhibition              | Booth, Sponsor, VenueMapNode                                                                                           | Booth belongs to Hall and ExhibitorProfile, and references an Asset for its logo. Sponsor references an Asset for its logo and carries a Tier. VenueMapNode marks a hall, a zone or a booth through Kind, PositionX and PositionY. Delegation is not a standalone entity: a delegate is a visitor with UserProfile.IsDelegate true and an invited Country, Country.IsInvited true. Two additive booleans, no new entity.                                                                                               |
| Engagement & Live       | SessionQuestion. The live state is held in columns on Session, not in an entity of its own.                            | SessionQuestion belongs to Session and is asked by a User. Questions close at session end and have no lower bound. Once the session is live, a hall carrying a geofence also requires a HallAttendance enter record.                                                                                                                                                                                                                                                                                                   |
| Networking              | Interest, backed by a Category of kind Interest; UserInterest, MeetingRequest, MatchSuggestion                         | UserInterest links User and an interest Category. MeetingRequest links a requester User and a target User. MatchSuggestion links a User to a suggested User; a score of 80 or more triggers a recommendation and a push.                                                                                                                                                                                                                                                                                               |
| Content & Media         | MediaItem, MediaPartner, NewsItem, Edition, EditionStat, EditionSpeaker                                                | Edition has many EditionStat and EditionSpeaker, and references cover and media Assets. NewsItem belongs to a news-category Category. MediaItem and MediaPartner reference Assets. The column sets of MediaItem and NewsItem are fixed per feature and not itemised here.                                                                                                                                                                                                                                              |
| Notifications           | Notification, NotificationDelivery                                                                                     | Notification belongs to a recipient User. NotificationDelivery belongs to Notification, one                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Context                | Entities                                                                     | Relationships                                                                                                                                                                                                                                                    |
|------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                                                                              | delivery row per channel: InApp, Email, SMS or WhatsApp.                                                                                                                                                                                                         |
| Cognitive AI           | FaqGroup, FaqEntry, AiSetting, SessionSummary                                | FaqGroup has many FaqEntry, across the two knowledge levels. SessionSummary belongs to Session. AiSetting is a key and value row.                                                                                                                                |
| Analytics & Statistics | GpsPresence, delivered as the table DevicePositionPings                      | GpsPresence belongs to User and holds batched, append-only telemetry. StatisticSnapshot is not built: dashboard figures are computed from the other contexts on each request.                                                                                    |
| Platform Configuration | Category, ContentBlock, VenueTrack, RegistrationControl, Asset, OperationLog | The single Category table carries every dynamic, client-maintained list, tagged by Kind and each with its own colour. Asset is referenced wherever a file is stored. OperationLog belongs to the acting User. RegistrationControl is a single configuration row. |

Those names are the logical model. Several delivered table names differ: AttendeeProfile is delivered as UserProfile, VenueEntry as the gate and HallAttendance model, Booking as SeatReservation, and MeetingRequest as SpeakerMeetingRequest, BusinessMeeting and DelegationMeetingRequest. Section 6.3 gives the mapping.

## 6.2 Entity Relationship Diagram

Two-database split. SIMF uses two physically separated SQL Server 2022 databases: SIMF\_Identity holds the Identity and Access tables through SimfIdentityDbContext, migration history \_\_EFMigrationsHistory\_Identity; SIMF\_App holds every other table through SimfAppDbContext, migration history \_\_EFMigrationsHistory\_App. Three consequences follow by design.

- No cross-database foreign keys. An App to Identity reference is a logical foreign key, a bare Guid enforced in application code, never a database constraint. Examples: UserProfile.UserId and GateScan.ScannedByUserId.
- No cross-database transaction. A unit of work touches one context and one database at a time.
- No duplicated data. Identity-owned data is resolved on read. The sole exception is the deliberate immutable audit-snapshot pattern, where an action log captures the actor's display name and email at write time.

The split is a security boundary: SIMF\_Identity can be backed up, encrypted and access-controlled independently of SIMF\_App. The two connection strings may point at one instance or at separate physical servers.

Core diagram. The diagram centres on User in SIMF\_Identity and on the programme, engagement and exhibition entities in SIMF\_App. Central keys: every entity has a GUID Id; User.Email and Badge.ReferenceNumber are unique; Permission is identified by its Code, in the format Page.Action; and the join tables UserRole, RolePermission, SessionSpeaker, UserInterest and NotificationDelivery carry composite links.

## SIMF core entity relationship diagram

Crow's foot notation. The two database frames are physically separate, so a relationship that crosses a frame is a logical reference held in application code, never a foreign key.

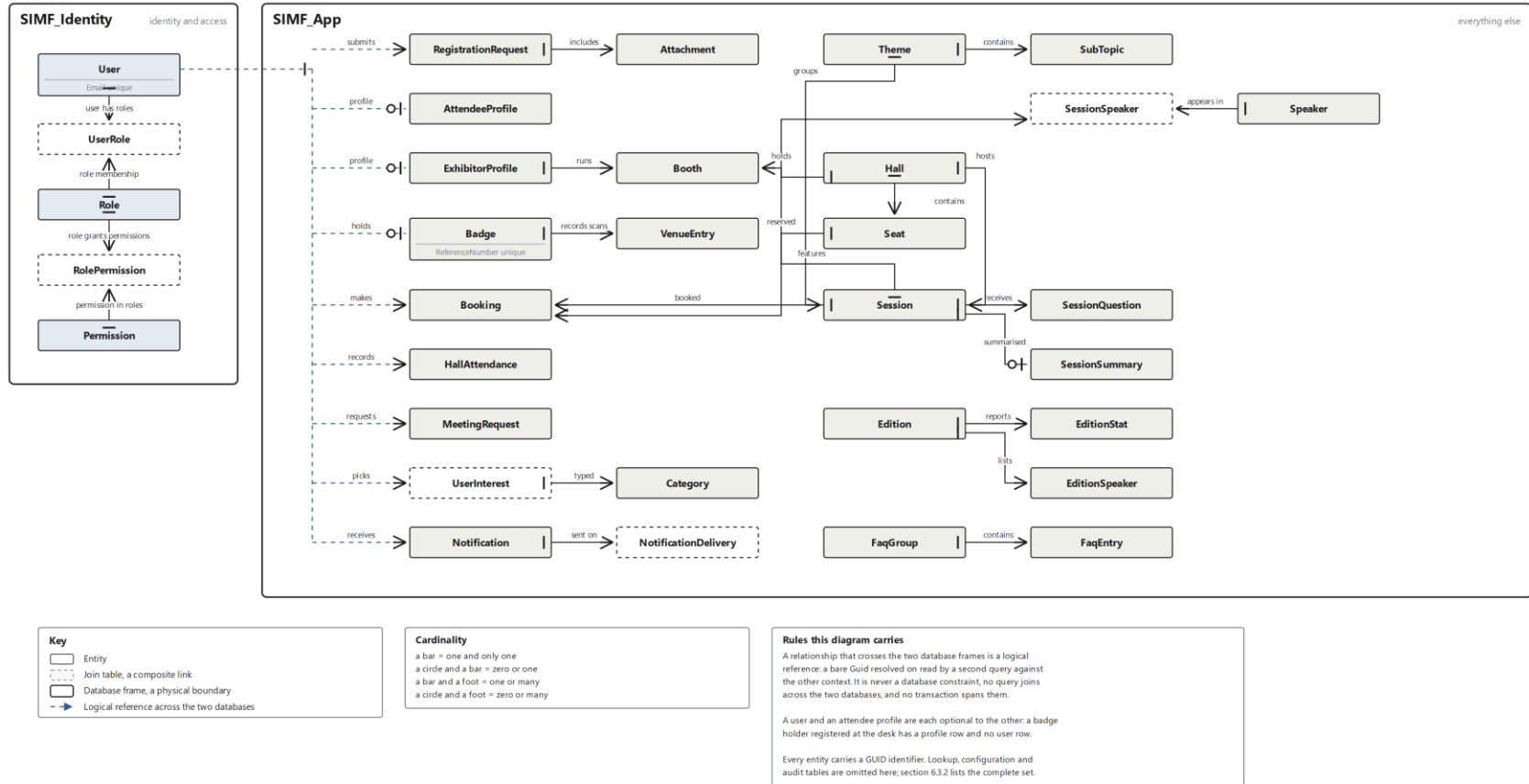


Figure 2. The core entity relationship diagram, in crow's foot notation. The two database frames are physically separate, so a relationship that crosses a frame is a logical reference held in application code and never a foreign key.

The table below indexes the diagram: each relationship drawn, its cardinality and its meaning.

| Entity              | Relates to                   | Cardinality  | Via                                                                            |
|---------------------|------------------------------|--------------|--------------------------------------------------------------------------------|
| User                | UserRole                     | 1 to many    | user has roles                                                                 |
| Role                | UserRole                     | 1 to many    | role membership                                                                |
| Role                | RolePermission               | 1 to many    | role grants perms                                                              |
| Permission          | RolePermission               | 1 to many    | perm in roles                                                                  |
| User                | AttendeeProfile              | 0..1 to 0..1 | user profile; a badge holder with no account has a profile row and no user row |
| User                | ExhibitorProfile             | 1 to 0..1    | exhibitor profile                                                              |
| User                | RegistrationRequest          | 1 to many    | submits                                                                        |
| RegistrationRequest | Attachment                   | 1 to many    | includes                                                                       |
| User                | Badge                        | 1 to 0..1    | holds                                                                          |
| Badge               | VenueEntry                   | 1 to many    | records scans                                                                  |
| User                | HallAttendance               | 1 to many    | records                                                                        |
| User                | Booking                      | 1 to many    | makes                                                                          |
| Theme               | SubTopic / Session           | 1 to many    | contains / groups                                                              |
| Hall                | Seat / Session               | 1 to many    | contains / hosts                                                               |
| Session             | SessionSpeaker               | 1 to many    | features                                                                       |
| Speaker             | SessionSpeaker               | 1 to many    | appears in                                                                     |
| Session             | Booking                      | 1 to many    | booked                                                                         |
| Seat                | Booking                      | 1 to many    | reserved                                                                       |
| Session             | SessionQuestion              | 1 to many    | receives                                                                       |
| Session             | SessionSummary               | 1 to 0..1    | summarised                                                                     |
| ExhibitorProfile    | Booth                        | 1 to many    | runs                                                                           |
| Hall                | Booth                        | 1 to many    | holds                                                                          |
| User                | MeetingRequest               | 1 to many    | requests                                                                       |
| User                | UserInterest                 | 1 to many    | picks                                                                          |
| Edition             | EditionStat / EditionSpeaker | 1 to many    | reports / lists                                                                |
| User                | Notification                 | 1 to many    | receives                                                                       |
| Notification        | NotificationDelivery         | 1 to many    | sent on                                                                        |
| FaqGroup            | FaqEntry                     | 1 to many    | contains                                                                       |
| Category            | UserInterest                 | 1 to many    | typed                                                                          |

The complete entity set, including the lookup, configuration and audit tables, is listed in section 6.3.2.

## 6.3 Schema Design

Two parts follow: a data dictionary for the tables carrying the most design weight, and an index of every logical entity.

### 6.3.1 Data dictionary, key tables

Type conventions for the tables below. GUID is the standard primary-key and foreign-key type. An enum(int) is an integer-backed set owned in C#. A datetime (Saudi local) is a Saudi local wall-clock timestamp. A string has its length fixed in the EF Core configuration and the migration, not reproduced here. The audit columns CreatedAt, CreatedBy, UpdatedAt, UpdatedBy, IsActive and DeletedAt apply through BaseAuditEntity and are not repeated per table except where load-bearing.

Users (SimfUser, deriving from IdentityUser of Guid), in SIMF\_Identity.

| Column       | Type   | Null | Key/Constraint      | Description                                         |
|--------------|--------|------|---------------------|-----------------------------------------------------|
| Id           | GUID   | No   | PK                  | User identifier                                     |
| Email        | string | No   | Unique (User.Email) | Login and contact email, from ASP.NET Core Identity |
| PasswordHash | string | Yes  | None                | From ASP.NET Core Identity                          |

| Column                 | Type                   | Null | Key/Constraint | Description                                                                                                                                                 |
|------------------------|------------------------|------|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SecurityStamp          | string                 | Yes  | None           | Identity stamp; drives immediate revoke                                                                                                                     |
| DisplayName            | string                 | No   | None           | Shown name                                                                                                                                                  |
| AccountState           | enum(int)              | No   | None           | Registered to Disabled; no separate IsActive on the user row                                                                                                |
| UserType               | enum(int)              | No   | None           | Visitor or Admin, as the enum name in nvarchar(16); integer 1 reserved                                                                                      |
| PasswordChangeRequired | bool                   | No   | None           | Forces a password change on next sign-in                                                                                                                    |
| PasswordChangedAt      | datetime (Saudi local) | Yes  | None           | Password max-age tracking                                                                                                                                   |
| LastSuccessfulSignInAt | datetime (Saudi local) | Yes  | None           | Dormant-account sweep input                                                                                                                                 |
| AvatarFileId           | string                 | Yes  | None           | The avatar's StoredFile row in the App database. Not a path, and deliberately not a database foreign key: the reference crosses the Identity and App split. |

Lockout and two-factor state are the standard ASP.NET Core Identity columns. The Identity tables carry no AspNet prefix: Users, Roles, UserRoles, UserClaims, UserLogins, UserTokens and RoleClaims.

Permissions (Permission), in SIMF\_Identity.

| Column | Type   | Null | Key/Constraint | Description                                                                                                                                                                  |
|--------|--------|------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Id     | GUID   | No   | PK             | Permission id                                                                                                                                                                |
| Code   | string | No   | Unique code    | Permission code, in the format Page.Action. With Id it is the whole table: the page, the action and the display name are read from PermissionCatalog in process, not stored. |

RolePermissions (RolePermission), in SIMF\_Identity.

| Column       | Type | Null | Key/Constraint             | Description     |
|--------------|------|------|----------------------------|-----------------|
| RoleId       | GUID | No   | PK part; FK to Roles       | Role side       |
| PermissionId | GUID | No   | PK part; FK to Permissions | Permission side |

Composite primary key (RoleId, PermissionId); a pure join table.

RefreshTokens (RefreshToken), in SIMF\_Identity.

| Column        | Type                   | Null | Key/Constraint                 | Description                                 |
|---------------|------------------------|------|--------------------------------|---------------------------------------------|
| Id            | GUID                   | No   | PK                             | Token record id                             |
| UserId        | GUID                   | No   | FK to the Identity Users table | Owning user                                 |
| TokenHash     | string                 | No   | None                           | Hashed token; the raw token is never stored |
| ExpiresAt     | datetime (Saudi local) | No   | None                           | Expiry                                      |
| RevokedAt     | datetime (Saudi local) | Yes  | None                           | Revocation time                             |
| RotatedFromId | GUID                   | Yes  | Self-reference                 | Rotation chain, for reuse detection         |

AccountCodes (AccountCode), in SIMF\_Identity.

| Column  | Type      | Null | Key/Constraint                                           | Description                                                      |
|---------|-----------|------|----------------------------------------------------------|------------------------------------------------------------------|
| Id      | GUID      | No   | PK                                                       | Code record id                                                   |
| UserId  | GUID      | Yes  | FK to the Identity Users table; CK_AccountCodes_OneOwner | Owning user. Null in the profile-owned case.                     |
| Purpose | enum(int) | No   | AccountCodePurpose                                       | EmailVerification, PasswordReset, SignInOtp, BadgeActivationOtp, |



| Column       | Type                   | Null | Key/Constraint | Description                                                                                                                                                                                                                                                                          |
|--------------|------------------------|------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                        |      |                | BiometricEnrolStepUp or EmailChangeVerification, persisted as the value name in nvarchar(32), not as an integer. A code belongs to a UserId or a UserProfileId, never both and never neither. BadgeActivationOtp is the profile-owned case, issued before the holder has an account. |
| Code         | string                 | No   | None           | Keyed HMAC of the one-time code; the code itself is emailed, never stored                                                                                                                                                                                                            |
| ExpiresAt    | datetime (Saudi local) | No   | None           | Expiry                                                                                                                                                                                                                                                                               |
| ConsumedAt   | datetime (Saudi local) | Yes  | None           | Redemption time                                                                                                                                                                                                                                                                      |
| AttemptCount | int                    | No   | None           | Per-code attempt cap                                                                                                                                                                                                                                                                 |

UserProfile, in SIMF\_App, the delivered form of AttendeeProfile. Bilingual.

| Column                                                 | Type      | Null | Key/Constraint                                                            | Description                                                                                                                                                                             |
|--------------------------------------------------------|-----------|------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Id                                                     | GUID      | No   | PK                                                                        | Profile id                                                                                                                                                                              |
| UserId                                                 | GUID      | Yes  | Filtered unique index over non-null rows                                  | Owning user. A logical link to the Identity Users table, no database foreign key, the reference crossing the database split. Null for a badge holder with a profile row and no account. |
| (bilingual name)                                       | string    | No   | None                                                                      | Arabic and English name, in paired columns                                                                                                                                              |
| Gender                                                 | enum(int) | Yes  | None                                                                      | Gender                                                                                                                                                                                  |
| Number, on the child table<br>ProfileIdentityDocuments | string    | Yes  | Encrypted at rest                                                         | Identity numbers. Personal data, encrypted through IPiiEncryptor.                                                                                                                       |
| (mobiles)                                              | string    | Yes  | Encrypted at rest                                                         | Mobile numbers. Personal data.                                                                                                                                                          |
| OrganisationId                                         | GUID      | Yes  | FK to Organisation                                                        | Organisation                                                                                                                                                                            |
| ProfileTypeId                                          | GUID      | Yes  | FK to ProfileTypes (FK_UserProfiles_ProfileTypes_ProfileTypeId, Restrict) | Profile type                                                                                                                                                                            |
| IsDelegate                                             | bool      | No   | None                                                                      | Delegation flag                                                                                                                                                                         |
| QrId                                                   | string    | Yes  | Filtered unique index over non-null rows                                  | Badge QR identifier, null until approval                                                                                                                                                |

| Column          | Type             | Null       | Key/Constraint | Description                                     |
|-----------------|------------------|------------|----------------|-------------------------------------------------|
|                 |                  |            |                | and minted when AdmissionState reaches Approved |
| ReferenceNumber | string           | No         | None           | SIMF reference number                           |
| (VIP columns)   | Group of columns | Not stated | Not stated     | VIP profile fields, not itemised here           |

Session, in SIMF\_App.

| Column                            | Type                   | Null       | Key/Constraint                           | Description                                                                                                                                                                            |
|-----------------------------------|------------------------|------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Id                                | GUID                   | No         | PK                                       | Session id                                                                                                                                                                             |
| Code                              | string                 | No         | None                                     | Session code                                                                                                                                                                           |
| HallId                            | GUID                   | No         | FK to Hall                               | Hosting hall                                                                                                                                                                           |
| CategoryId                        | GUID                   | Yes        | FK to SessionCategory                    | Session category, a dynamic client-maintained list                                                                                                                                     |
| Type                              | enum(int)              | Yes        | SessionType                              | Session type                                                                                                                                                                           |
| Start / End                       | datetime (Saudi local) | No         | None                                     | Session start and end                                                                                                                                                                  |
| Status                            | enum(int)              | No         | SessionStatus                            | Scheduled, Held, Recorded or Published                                                                                                                                                 |
| LiveStreamFileId                  | string                 | Yes        | FK_Sessions_StoredFiles_LiveStreamFileId | The StoredFiles row holding the YouTube live-stream URL                                                                                                                                |
| LiveSignLanguageFileId            | string                 | Yes        | Reference to StoredFiles                 | The StoredFiles row holding the sign-language feed URL. The feature is withdrawn from the delivered scope; the column and the Control Panel session field that writes it are retained. |
| LiveCaptions / LiveCaptionsArabic | string                 | Yes        | None                                     | Live captions, English and Arabic                                                                                                                                                      |
| RecordingFileId                   | Not stated             | Not stated | Not stated                               | Recording metadata                                                                                                                                                                     |
| ReminderSentAt                    | datetime (Saudi local) | Yes        | None                                     | Session-reminder dedup guard                                                                                                                                                           |

SeatReservation, in SIMF\_App, the delivered form of Booking.

| Column                | Type            | Null | Key/Constraint                 | Description                                                          |
|-----------------------|-----------------|------|--------------------------------|----------------------------------------------------------------------|
| Id                    | GUID            | No   | PK                             | Reservation id                                                       |
| SessionId             | GUID            | No   | FK to Session                  | Session booked                                                       |
| RowLabel / SeatNumber | string(8) / int | Yes  | Seat position on the hall grid | Seat. Both are null for open seating.                                |
| Kind                  | enum(int)       | No   | SeatReservationKind            | UserBooking=0, AdminReservedRow=1, RandomAssignment=2, OpenSeating=3 |
| Status                | enum(int)       | No   | BookingStatus                  | Approved on every create path, Cancelled on release.                 |

| Column | Type | Null | Key/Constraint | Description                                                                                                                                 |
|--------|------|------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
|        |      |      |                | Pending and Rejected remain in the enum, but no path writes them and no approve or reject action exists on the API or in the Control Panel. |

A filtered unique index allows one active reservation per seat per session.

GateScan, in SIMF\_App. An append-only audit log.

| Column                | Type                      | Null | Key/Constraint                              | Description                                                                         |
|-----------------------|---------------------------|------|---------------------------------------------|-------------------------------------------------------------------------------------|
| Id                    | bigint<br>IDENTITY        | No   | Clustered PK                                | Monotonic, so the clustered index does not fragment                                 |
| GateId                | GUID                      | No   | FK to Gate                                  | Gate scanned at                                                                     |
| UserProfileId         | GUID                      | Yes  | Logical FK to UserProfile                   | Scanned visitor                                                                     |
| QrIdAtScan            | string(96)                | No   | None                                        | QR id preserved exactly, kept after rotation                                        |
| Direction             | enum(int)                 | No   | CheckIn / CheckOut                          | Scan direction                                                                      |
| Outcome               | enum(int)                 | No   | Allowed / Denied                            | Scan outcome                                                                        |
| DenialReasonCode      | string                    | Yes  | CK_GateScans_DenialPin                      | Denial reason, an integer enum 0 to 8. Required on a denial, forbidden on an allow. |
| ScannedAt             | datetime<br>(Saudi local) | No   | Indexed                                     | Server clock, the authoritative scan time                                           |
| ClientScannedAt       | datetime<br>(Saudi local) | Yes  | None                                        | Device clock, asserted by the client                                                |
| ScannedByUserId       | GUID                      | No   | Logical FK to SimfUser                      | Operator who scanned                                                                |
| Source                | enum(int)                 | No   | Simulator / MobileApp / Kiosk               | Scan source                                                                         |
| CorrelationId         | string                    | No   | None                                        | Request correlation                                                                 |
| IpAddress / UserAgent | string                    | Yes  | None                                        | Client metadata                                                                     |
| IdempotencyKey        | string                    | Yes  | Unique filtered<br>(IdempotencyKey, GateId) | Replay enforcement                                                                  |

Append-only is enforced at the database as well as in the application: the runtime account may insert, and may not update or delete. Scan records are also shipped to the ministry log collector. The table opts out of RowAudit.

Notifications (Notification), in SIMF\_Identity.

| Column                              | Type                      | Null | Key/Constraint                                                                 | Description                                                      |
|-------------------------------------|---------------------------|------|--------------------------------------------------------------------------------|------------------------------------------------------------------|
| Id                                  | GUID                      | No   | PK                                                                             | Notification id                                                  |
| UserId                              | GUID                      | No   | FK to the Identity Users table                                                 | Recipient                                                        |
| Kind                                | enum(int)                 | No   | NotificationKind, stored as its name in nvarchar(64) rather than as an integer | For example BookingConfirmed, SessionReminder or BookingRejected |
| Title / TitleArabic                 | string                    | No   | None                                                                           | Bilingual title                                                  |
| Body / BodyArabic                   | string                    | Yes  | None                                                                           | Bilingual body                                                   |
| Severity                            | enum(int)                 | No   | NotificationSeverity                                                           | Info, Success, Warning or Error                                  |
| ReadAt                              | datetime<br>(Saudi local) | Yes  | None                                                                           | Read time                                                        |
| RelatedEntityType / RelatedEntityId | string / GUID             | Yes  | None                                                                           | Deep-link target                                                 |

The logical model separates Notification from NotificationDelivery, one row per delivery channel. The delivered Notification table sits in SIMF\_Identity; the per-channel delivery model is retained in the logical model.

### 6.3.2 Complete entity index

Every entity of the logical data model, by context; the list is exhaustive. Section 6.3.1 records the delivered name where it differs.

| Context                 | Entities                                                                                                               |
|-------------------------|------------------------------------------------------------------------------------------------------------------------|
| Identity & Access       | User, Role, Permission, RolePermission, UserRole, RefreshToken, EmailVerificationCode, TotpSecret                      |
| Registration & Approval | RegistrationRequest, AttendeeProfile, ExhibitorProfile, Companion, Attachment                                          |
| Badge & Access Control  | Badge, VenueEntry, HallAttendance, SavedContact, Gate, GateProfileTypeAllow, GateAssignment, GateScan, ScanIdempotency |
| Forum Programme         | Theme, SubTopic, Hall, Seat, Session, Speaker, SessionSpeaker, SpeakerPresentation, Booking                            |
| Exhibition              | Booth, Sponsor, VenueMapNode                                                                                           |
| Engagement & Live       | SessionQuestion                                                                                                        |
| Networking              | Interest, UserInterest, MeetingRequest, MatchSuggestion                                                                |
| Content & Media         | MediaItem, MediaPartner, NewsItem, Edition, EditionStat, EditionSpeaker                                                |
| Notifications           | Notification, NotificationDelivery                                                                                     |
| Cognitive AI            | FaqGroup, FaqEntry, AiSetting, SessionSummary                                                                          |
| Analytics & Statistics  | DevicePositionPing                                                                                                     |
| Platform Configuration  | Category, ContentBlock, VenueTrack, RegistrationControl, Asset, OperationLog                                           |

Additions in the delivered schema. SIMF\_App carries further additive entities, including ProgrammeDay, SessionTheme, SessionCategory, SessionModerator, HallSeatLayout, Exhibitor, ExhibitorMembership, ExhibitorVisitorScan, ArchiveEdition with ArchiveMediaItem, ArchiveSessionTitle and ArchivePastSpeaker, RatingType with RatingQuestionGroup, RatingQuestion, RatingResponse and RatingAnswer, Banner, Connection, Invitation, MeetingTable, HallAllocation, BusinessMeeting with BusinessMeetingParticipant, SpeakerMeetingRequest, SpeakerAvailabilityWindow, DelegationMeetingRequest, SavedContact, ContactInquiry, VisitorShareToken, SystemSetting, OrganizationProfile with OrganizationAboutItem and OrganizationDetail, AiPrompt, AiPromptHistory, AiInvocation and RowAudit. SIMF\_Identity carries SecondFactorToken, TotpRecoveryCode, DeviceKey and PasswordHistoryEntry. AccountCode supersedes EmailVerificationCode, and TotpSecret is absorbed into the Identity token store.

Authoritative full schema. The per-column schema for all tables, approximately 100 across the two databases, is defined by one InitialCreate migration per context with the EF Core entity configurations. Exact column types, lengths and index definitions are reviewed as code there; columns not in section 6.3.1 are not reproduced.

## 7. Solution Description

This section covers the layered logical view, the physical deployment and its node specifications, the technology stack, folder layouts and project dependencies, the patterns applied across every feature context, and an inventory of every third-party package.

## 7.1 Logical View

SIMF is a modular monolith with Domain-Driven Design layering. The backend is a single deployable API host of feature-aligned bounded contexts under a strictly inward dependency rule: SIMF.Api depends on SIMF.Infrastructure, SIMF.Infrastructure on SIMF.Application, SIMF.Application on SIMF.Domain. Nothing points outward, which keeps the domain testable and confines persistence to one layer. Two web front-ends, the Control Panel and the Website, and one Flutter mobile application consume the API; both web front-ends apply a Backend-for-Frontend token-handling pattern, so a browser never holds a raw bearer token.

## SIMF layered solution structure

A modular monolith with domain-driven layering. Each layer encloses the layer it depends on, so a dependency can only point inward.

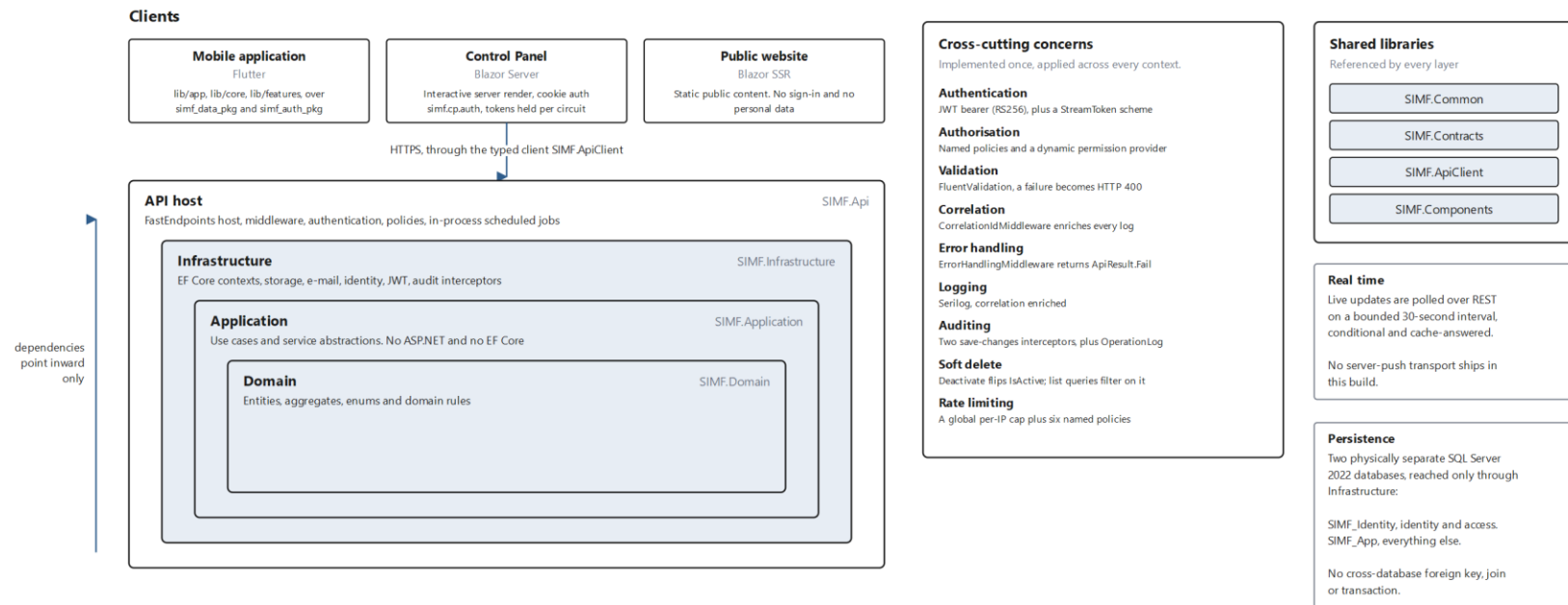


Figure 3. The layered solution structure. Each layer encloses the layer it depends on, so a dependency can only point inward, and the domain compiles with no knowledge of the host, the database or any client.

The platform is deployed on-premises across two security areas, SSA and HSA. SSA holds three zones. Access: the attendee mobile device, the public website browser and the administrator browser. Perimeter: the web application firewall and the load balancer, where TLS terminates. Presentation: the mobile edge (SIMF.MobileEdge), the website (SIMF.Web) and the Control Panel (SIMF.ControlPanel). HSA is a single zone, holding the API load balancer and four API nodes (SIMF.Api), the LLM server running GPT OSS 120B and the mail relay, both provided by SITE with the service, two database nodes carrying SIMF\_Identity and SIMF\_App, and MinIO object storage. No firewall sits inside HSA; the internal firewall at its boundary is the only route in, admitting nothing but HTTPS 443 from the presentation zone to the API. No host in HSA has a route to the internet.

| Node                                                            | Count | Specification                                                                 |
|-----------------------------------------------------------------|-------|-------------------------------------------------------------------------------|
| Website (SIMF.Web)                                              | 2     | 32 GB RAM, 16 vCPU, 300 GB storage, Windows Server 2022                       |
| Control Panel (SIMF.ControlPanel)                               | 2     | 32 GB RAM, 16 vCPU, 300 GB storage, Windows Server 2022                       |
| Mobile edge (SIMF.MobileEdge)                                   | 2     | 16 GB RAM, 8 vCPU, 100 GB storage; a proposed minimum confirmed with the site |
| API (SIMF.Api)                                                  | 4     | As the website and Control Panel nodes                                        |
| Database (SIMF_Identity and SIMF_App)                           | 2     | 64 GB RAM, 8 vCPU, 2 TB storage                                               |
| Object storage (MinIO)                                          | 2     | 16 GB RAM, 8 vCPU, 4 TB storage per node                                      |
| Caller                                                          |       | Protocol and port                                                             |
| Mobile edge, website and Control Panel in the presentation zone |       | API (SIMF.Api) in HSA, across the internal firewall                           |
| API (SIMF.Api)                                                  |       | LLM server running GPT OSS 120B                                               |
| API (SIMF.Api)                                                  |       | Mail relay                                                                    |
| API (SIMF.Api)                                                  |       | Databases SIMF_Identity and SIMF_App                                          |
| API (SIMF.Api)                                                  |       | MinIO object storage, S3 API                                                  |
| Control Panel (SIMF.ControlPanel)                               |       | YouTube caption API, the single outbound call from the platform               |

### 7.1.1 Backend layers

| Layer          | Project or mechanism                                                 | Responsibility                                                                                                                                              |
|----------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Domain         | SIMF.Domain                                                          | Entities, aggregates, enums and domain rules. One framework package, Microsoft.Extensions.Identity.Stores, because SimfUser derives from IdentityUser<Guid> |
| Application    | SIMF.Application                                                     | Use cases and service abstractions; no ASP.NET Core, no EF Core                                                                                             |
| Infrastructure | SIMF.Infrastructure                                                  | EF Core contexts, storage, e-mail, identity, JWT and the audit interceptors                                                                                 |
| API host       | SIMF.Api                                                             | FastEndpoints host, middleware, authentication, policies and in-process scheduled jobs                                                                      |
| Real-time      | REST polling on a 30-second interval, conditional and cache-answered | Live updates polled over REST on a bounded interval. No server-push transport ships in this build                                                           |

### 7.1.2 Client and shared layers

- Control Panel: a Backend-for-Frontend on Blazor Server, rendered interactive-server with prerendering disabled. It authenticates with the cookie simf.cp.auth, holds tokens per circuit and forwards them to the API through the typed client.
- Website: Blazor SSR for static public content. With no sign-in, no registration, no account page and no personal data stored, it carries no authentication surface and no per-page permissions. Its one

credentialed entry point is the meeting-confirmation page, accepting an opaque per-speaker token carried in the link.

- Flutter mobile application: lib/app for the shell, theme, localisation and shared widgets; lib/core for environment, networking, session, responsive layout, validation and utilities; lib/features for the per-feature screens. It sits over two local packages, simf\_data\_pkg for the HTTP layer and simf\_auth\_pkg for authentication, session and device keys.
- Shared libraries: SIMF.Common, SIMF.Contracts, SIMF.ApiClient and SIMF.Components.

### 7.1.3 Cross-cutting concerns

Each concern is implemented once and applied across every feature context.

| Concern        | Mechanism                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------|
| Authentication | JWT bearer signed with RS256 as the default scheme, plus a distinct StreamToken scheme                       |
| Authorisation  | Named policies and a dynamic permission-policy provider over PermissionCatalog                               |
| Validation     | FluentValidation, through Validator<TRequest>. A failure returns HTTP 400 with the code VALIDATION_FAILED    |
| Correlation    | CorrelationIdMiddleware reads or generates X-Correlation-Id and enriches the logs                            |
| Error handling | ErrorHandlingMiddleware converts an exception to ApiResult.Fail                                              |
| Logging        | Serilog, enriched with the correlation identifier                                                            |
| Auditing       | AuditStampingSaveChangesInterceptor and RowAuditingSaveChangesInterceptor, plus an explicit OperationLog     |
| Soft-delete    | BaseAuditEntity.Deactivate() flips IsActive; list queries filter on IsActive equal to true                   |
| Rate limiting  | A global per-IP cap plus six named policies: auth, auth-email, ai-test, ai-assistant, operational and lookup |

## 7.2 Solution Structure

### 7.2.1 Technology Stack

Every version below is read from the repository; a technology with no pinned version is marked as such.

| Layer             | Technology                                                       | Version                                                                                                                        | Purpose                                                       |
|-------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Runtime           | .NET                                                             | 10.0, packages 10.0.x                                                                                                          | Backend runtime and framework                                 |
| API framework     | FastEndpoints                                                    | 8.1.0                                                                                                                          | Endpoint host and routing                                     |
| API documentation | FastEndpoints.Swagger                                            | 8.1.0                                                                                                                          | OpenAPI and Swagger generation                                |
| Data access       | EF Core, SQL Server provider                                     | 10.0.8                                                                                                                         | Object-relational mapping and migrations                      |
| Database          | SQL Server                                                       | 2022. Standard edition in development and test, Enterprise edition in an AlwaysOn Availability Group in staging and production | Relational store, two databases                               |
| Control Panel     | Blazor Server                                                    | ASP.NET Core 10, no separate pinned version                                                                                    | Administrative user interface                                 |
| Website           | Blazor SSR                                                       | ASP.NET Core 10, no separate pinned version                                                                                    | Public site                                                   |
| Component library | SIMF.Components, the Simf* components, with Cropper.Blazor 1.5.1 | 10.0.7 for Components.Web                                                                                                      | Shared Control Panel and Website components and design tokens |
| Mobile            | Flutter and Dart                                                 | Flutter 3.27.0 or later, Dart SDK 3.6.0 or later                                                                               | Android and iOS application, from the pubspec environment     |



| Layer   | Technology                 | Version | Purpose                          |
|---------|----------------------------|---------|----------------------------------|
|         |                            |         | block                            |
| Logging | Serilog.AspNetCore         | 10.0.0  | Structured logging               |
| E-mail  | MailKit                    | 4.16.0  | SMTP send                        |
| Hosting | IIS on Windows Server 2022 | 10.x    | Reverse-proxied on-premises host |

The Control Panel and Website user interface rests on the shared Simf\* Blazor component library, SIMF.Components, with theme.tokens.css. Cropper.Blazor supplies image cropping.

### 7.2.2 Backend Folder Structure

The backend and shared projects sit under src/, from the repository project list and folder inventory.

| Project path                    | Contents                                                                     |
|---------------------------------|------------------------------------------------------------------------------|
| src/Backend/SIMF.Domain         | Entities, aggregates, enums and domain rules, with no framework dependencies |
| src/Backend/SIMF.Application    | Use cases and service abstractions; no ASP.NET Core, no EF Core              |
| src/Backend/SIMF.Infrastructure | EF Core contexts, storage, e-mail, identity, JWT and audit                   |
| src/Backend/SIMF.Api            | FastEndpoints host                                                           |
| src/Shared/SIMF.Common          | ApiResponse<T>, PermissionCatalog, AppRoles, enums and GridQuery             |
| src/Shared/SIMF.Contracts       | Request and response data transfer objects                                   |
| src/Shared/SIMF.ApiClient       | Typed HTTP client for the Control Panel and the Website                      |
| src/Shared/SIMF.Components      | Shared Simf* Blazor components and theme tokens                              |

SIMF.Api holds these host-level folders.

- App\_Data
- Authentication: JWT and StreamToken scheme setup
- Authorization: PermissionAuthorization, the policy provider and handler
- Endpoints: feature-grouped endpoint classes
- HostedServices: in-process scheduled jobs, among them the dormant sweep and reminders
- Middleware: correlation, security headers, error handling, the e-mail rate key and Swagger authentication
- RateLimiting
- RequestContext: HttpRequestContext, implementing IRequestContext
- logs

SIMF.Api/Endpoints/ has one folder per feature, aligned to the bounded contexts: Account, Admin, Ai, Archive, Assets, Attendance, Auth, BusinessMeetings, Contacts, Exhibitors, Feedback, Files, Gates, Networking, News, Operations, Programme, Public, PublicRelations, Recommendations, Reporting, Requests, Sessions, Sponsors, Staff, Statistics and Support.

SIMF.Infrastructure carries one folder per feature or service: AccessControl, Ai, Archive, Assets, Attendance, Auditing, BusinessMeetings, Cms, Common, Configuration, Contacts, Delegations, Editions, Email, Excel, Exhibition, Exhibitors, Faq, Feedback, Files, Identity, IdentityAccess, Logs, Media, MeetingRequests, MyArea, Networking, Notifications, Operations, Organisations, Persistence, Preferences, Programme, PublicRelations, Recommendations, Regions, Reporting, Requests, SeatReservations, Seeding, SessionQuestions, Sponsors, Statistics, Support and Venue.

### 7.2.3 Frontend Folder Structure

The Flutter application lives at src/Mobile/simf\_app, its lib/ folder in three groups.

- lib/app holds localization for the bilingual Arabic and English resources, theme for the navy-always theme, and widgets for the global app-bar invariant and shared widgets.
- lib/core holds env for the environment and API base address, errors, net for the networking glue, organization\_profile, responsive for the flexible-width helpers, session, sharing, site\_settings, startup for the cold-start authentication restore and splash gate, utils, validation and widgets.
- lib/features holds one folder per feature area.

The feature folders are about, accessibility, account, ai\_summary, archive, badge, booths, chatbot, contact\_us, contacts, content, delegations, exhibition, exhibitor, faq, feedback, forum\_guide, gallery, gates, guest, home, live, media\_partners, meet, meetings, moderation, more, myarea, news, notifications, onboarding, questions, registration, requests, sessions, speakers, splash, sponsors, staff and venuemap.

Two local packages sit under src/Mobile/simf\_app/packages/. simf\_data\_pkg is the single HTTP layer, holding SimfApiClient, ApiResult, ApiFailure and the secure and preferences storage. simf\_auth\_pkg holds the authentication controller and state, the session, AppRole and the biometric ES256 device keys.

For the Blazor clients the authoritative layout is the projects themselves, src/ControlPanel/SIMF.ControlPanel and src/Website/SIMF.Web. The detail below is representative, not exhaustive.

- src/ControlPanel/SIMF.ControlPanel: Endpoints/ holds AuthEndpoints.cs, the Backend-for-Frontend credential and second-factor exchange issuing the cookie. Authorization/ holds PermissionAuthorization.cs, the policy provider and its handler. CpNavigation.cs declares 14 navigation groups over the /admin/\* routes. The remaining folders hold the pages, built on SimfDataGrid and the shared CRUD framework.
- src/Website/SIMF.Web: the server-rendered public pages and their interactive islands.

## 7.2.4 Project Dependencies

The project-reference graph follows the layering's inward rule: an outer project references an inner one, never the reverse.

| Project             | References                                                         | Direction note                                                         |
|---------------------|--------------------------------------------------------------------|------------------------------------------------------------------------|
| SIMF.Domain         | SIMF.Common                                                        | Innermost. One framework package, Microsoft.Extensions.Identity.Stores |
| SIMF.Application    | SIMF.Domain, SIMF.Common, SIMF.Contracts                           | Inward to Domain                                                       |
| SIMF.Infrastructure | SIMF.Domain, SIMF.Application                                      | Inward to Application and Domain                                       |
| SIMF.Api            | SIMF.Application, SIMF.Infrastructure, SIMF.Common, SIMF.Contracts | Host, composes all layers                                              |
| SIMF.Common         | None                                                               | Shared kernel, references nothing                                      |
| SIMF.Contracts      | SIMF.Common                                                        | Data transfer objects                                                  |
| SIMF.ApiClient      | SIMF.Common, SIMF.Contracts                                        | Typed client for the Control Panel and the Website                     |
| SIMF.Components     | SIMF.Common, Microsoft.AspNetCore.Components.Web                   | Shared Blazor components and theme tokens                              |
| SIMF.ControlPanel   | SIMF.ApiClient, SIMF.Common, SIMF.Contracts, SIMF.Components       | Blazor Server administrative Backend-for-Frontend                      |
| SIMF.Web            | SIMF.ApiClient, SIMF.Common, SIMF.Contracts, SIMF.Components       | Blazor SSR public site and Backend-for-Frontend                        |

The Control Panel and the Website reference only the shared client, contract and component projects, never the backend, and so reach the API server to server through the typed client.

### 7.2.5 Patterns & Practices

These patterns apply across the backend.

- FastEndpoints endpoint pattern. An endpoint is a class with `Configure()`, declaring route, verbs, policies or anonymous access, tags, rate-limiting policy and summary, and `HandleAsync()`, carrying the behaviour. Application routes sit under `/api/v1/app/*`, administrative routes under `/api/v1/admin/*`.
- `ApiResponse<T>` envelope. Every response carries Success, Data, Error and Meta. Error is an `ApiError` with Code, Message, MessageArabic and a list of `ApiErrorDetail`; Meta carries pagination and counts. `ApiCallResult<T>` pairs an HTTP status with the envelope.
- FluentValidation triple-lock. A field's maximum length is held identical in the FluentValidation rule, the EF Core `HasMaxLength` configuration and the user-interface `MaxLength`. A failure converts to HTTP 400 with the code `VALIDATION_FAILED` and bilingual field detail.
- Permission catalogue. `PermissionCatalog`, at `src/Shared/SIMF.Common/PermissionCatalog.cs`, is the single source of truth. It sets Wildcard to `"*"`, ClaimType to `"perm"` and PolicyPrefix to `"perm:"`, nesting codes as `Page.Action`. API and Control Panel both read it; tests fail the build when a page or endpoint is left ungated.
- Repository and service abstractions. Use-case services live in `SIMF.Application` behind interfaces, among them `IJwtTokenService`, `IRequestContext` and the central file-store abstraction `IFileService`; implementations live in `SIMF.Infrastructure`.
- Soft-delete through `BaseAuditEntity.Deactivate()`, as in section 7.1.3.
- Audit interceptors. `AuditStampingSaveChangesInterceptor` writes created and updated stamps; `RowAuditingSaveChangesInterceptor` writes the row-level change log. Both run on the App context; the Identity context runs the row-audit interceptor. Both audit tables are append-only: the application never updates or deletes an audit row, the runtime database account may insert but not update or delete, and records are also shipped to the ministry log collector.
- Options pattern. Typed configuration sections bind through it. A production override arrives as a per-host prefixed environment variable with double-underscore nesting: `SIMF_API_` for the API, `SIMF_CP_` for the Control Panel, `SIMF_WEB_` for the Website, `SIMF_EDGE_` for the mobile edge. Secrets are not among them: the token signing key and the two data-encryption keys are read at start-up from the central key management service, backed by a hardware security module, and a boot guard refuses to start a node whose secrets cannot be read.
- Guard clauses and deterministic failure. `DataValidationException`, extending `ApiException`, signals a validation failure, and the error middleware maps each exception to its status code.
- Pagination. Paged application GET endpoints take `skip` and `top` parameters, except GET `/app/news`, which takes `page` and `pageSize`. Administrative grids POST a `GridQuery` body and receive a `GridPage<T>`. Each grid declares its own default page size and maximum, applied by `GridQuery.ClampPage`.

Naming, error-handling, logging and security conventions follow from the same layering. Feature-folder organisation is mirrored across Endpoints, Application and Infrastructure. Bilingual content carries `Name` and `NameArabic` fields. Timestamps are presented in Saudi local time. The correlation identifier propagates through the logs and audit rows. CORS uses an explicit allow-list, never a wildcard. In production Swagger is gated behind Basic authentication and a feature flag, with a boot guard.

### 7.2.6 Packages Inventory

The NuGet packages and versions below are read from the repository, deduplicated across projects and grouped by category.

| Category                                | Package                                              | Version  | Purpose                                    |
|-----------------------------------------|------------------------------------------------------|----------|--------------------------------------------|
| Framework and runtime                   | Microsoft.AspNetCore.Components.Web                  | 10.0.7   | Blazor component primitives                |
| Framework and runtime                   | Microsoft.Extensions.Configuration.Abstractions      | 10.0.8   | Configuration abstractions                 |
| Framework and runtime                   | Microsoft.Extensions.Hosting.Abstractions            | 10.0.8   | Hosting abstractions                       |
| Framework and runtime                   | Microsoft.Extensions.Options.ConfigurationExtensions | 10.0.8   | Options-pattern binding                    |
| Framework and runtime                   | Microsoft.Extensions.Logging.Abstractions            | 10.0.8   | Logging abstractions                       |
| Data access                             | Microsoft.EntityFrameworkCore.SqlServer              | 10.0.8   | EF Core SQL Server provider                |
| Data access                             | Microsoft.EntityFrameworkCore.Design                 | 10.0.8   | EF migrations and design-time tooling      |
| Authentication and security             | Microsoft.AspNetCore.Authentication.JwtBearer        | 10.0.8   | JWT bearer authentication                  |
| Authentication and security             | Microsoft.AspNetCore.Identity.EntityFrameworkCore    | 10.0.8   | ASP.NET Identity stores                    |
| Authentication and security             | Microsoft.Extensions.Identity.Stores                 | 10.0.8   | Identity store primitives                  |
| Authentication and security             | System.IdentityModel.Tokens.Jwt                      | 8.18.0   | JWT token creation and validation          |
| Authentication and security             | Otp.NET                                              | 1.4.1    | TOTP second factor                         |
| API and serialisation                   | FastEndpoints                                        | 8.1.0    | Endpoint framework                         |
| API and serialisation                   | FastEndpoints.Swagger                                | 8.1.0    | OpenAPI and Swagger documents              |
| Logging                                 | Serilog.AspNetCore                                   | 10.0.0   | Structured logging integration             |
| Logging                                 | Serilog.Sinks.File                                   | 7.0.0    | File log sink                              |
| Imaging, QR and machine learning        | QRCode                                               | 1.6.0    | QR code generation                         |
| Imaging, QR and machine learning        | Cropper.Blazor                                       | 1.5.1    | Image cropping in the Control Panel        |
| Imaging, QR and machine learning        | FaceAiSharp.Bundle                                   | 0.6.35   | Face processing for identity and liveness  |
| Imaging, QR and machine learning        | Microsoft.ML.OnnxRuntime                             | 1.26.0   | ONNX model inference                       |
| E-mail, office and other infrastructure | MailKit                                              | 4.16.0   | SMTP e-mail send                           |
| E-mail, office and other infrastructure | ClosedXML                                            | 0.104.2  | Excel import and export                    |
| Documents, gateway and components       | QuestPDF                                             | 2026.7.1 | Badge PDF generation                       |
| Documents, gateway and components       | Yarp.ReverseProxy                                    | 2.3.0    | Mobile edge reverse proxy                  |
| Documents, gateway and components       | Microsoft.AspNetCore.Components.Authorization        | 10.0.7   | Referenced by the shared component library |

The Flutter packages below are read from the pubspec, with its caret constraints.

| Package                     | Version       | Purpose                                                            |
|-----------------------------|---------------|--------------------------------------------------------------------|
| flutter_riverpod            | ^3.4.2        | State management                                                   |
| go_router                   | ^14.6.0       | Routing and shell navigation                                       |
| intl                        | ^0.19.0       | Internationalisation and formatting                                |
| flutter_localizations       | From the SDK  | Localisation delegates                                             |
| local_auth                  | ^2.3.0        | Biometric authentication                                           |
| image_picker                | ^1.1.2        | Image selection                                                    |
| google_mlkit_face_detection | ^0.13.1       | Face detection for liveness                                        |
| camera                      | ^0.11.0+2     | Camera capture                                                     |
| share_plus                  | ^10.0.0       | Native share                                                       |
| add_2_calendar              | ^3.0.1        | Adds a session to the device calendar                              |
| qr_flutter                  | ^4.1.0        | QR rendering for the badge                                         |
| video_player                | ^2.9.1        | Video playback, the HLS and MP4 fallback                           |
| youtube_player_iframe       | ^6.0.2        | Live YouTube feed                                                  |
| url_launcher                | ^6.3.0        | External links                                                     |
| flutter_zxing               | ^2.3.0        | QR and barcode scanning                                            |
| flutter_svg                 | ^2.0.10       | SVG rendering                                                      |
| package_info_plus           | ^8.3.0        | Application version information                                    |
| pub_semver                  | ^2.1.4        | Version comparison                                                 |
| pointycastle                | ^3.9.1        | AES-256-GCM offline badge decode and ES256 device-key cryptography |
| simf_data_pkg               | Local package | HTTP and data layer                                                |
| simf_auth_pkg               | Local package | Authentication and session                                         |

## 8. Requirement Traceability

The Business Requirements Document delivered with this design, SIMF-BRD-001, carries 132 numbered requirements: 14 business rules, 90 functional requirements, 12 non-functional requirements and 16 operational requirements. This section maps every one of those identifiers to the design content that implements it and to the method by which it is verified. Each identifier holds its own row. A reviewer can therefore search for a single requirement and follow it from the business document into this design, and back again, without expanding a range.

The verification method is fixed by the kind of requirement.

- A functional requirement is verified by test.
- A business rule is verified by test.
- A non-functional requirement is verified by test and analysis.
- An operational requirement is verified by demonstration.
- A withdrawn requirement has no verification method. Its row records that verification is not applicable.

Four of the 132 requirements are withdrawn: FR-706, FR-707, FR-807 and BR-08. The Business Requirements Document marks them withdrawn. Nothing in this design implements them and no test covers them. Their rows are kept so that the total of 132 reconciles against the business document.

## 8.1 Functional Requirements

The Service column names the service that implements the requirement. The Designed in column names the section of this document that specifies it. FR-804 is designed in 5.8 and 5.13.

| Identifier | Service                          | Designed in                   | Verification method |
|------------|----------------------------------|-------------------------------|---------------------|
| FR-101     | F-01 Accounts and Authentication | 5.1 Authentication and Login  | Test                |
| FR-102     | F-01 Accounts and Authentication | 5.1 Authentication and Login  | Test                |
| FR-103     | F-01 Accounts and Authentication | 5.1 Authentication and Login  | Test                |
| FR-104     | F-01 Accounts and Authentication | 5.1 Authentication and Login  | Test                |
| FR-105     | F-01 Accounts and Authentication | 5.1 Authentication and Login  | Test                |
| FR-106     | F-01 Accounts and Authentication | 5.1 Authentication and Login  | Test                |
| FR-107     | F-01 Accounts and Authentication | 5.1 Authentication and Login  | Test                |
| FR-108     | F-01 Accounts and Authentication | 5.1 Authentication and Login  | Test                |
| FR-201     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-202     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-203     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-204     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-205     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-206     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-207     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-208     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-209     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-210     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-211     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-212     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-213     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-214     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-215     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-216     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |
| FR-217     | F-02 Registration and Approval   | 5.2 Registration and Approval | Test                |

| Identifier | Service                           | Designed in                             | Verification method       |
|------------|-----------------------------------|-----------------------------------------|---------------------------|
| FR-301     | F-03 Badge and Access Control     | 5.3 Badge and Access Control, Gates     | Test                      |
| FR-302     | F-03 Badge and Access Control     | 5.3 Badge and Access Control, Gates     | Test                      |
| FR-303     | F-03 Badge and Access Control     | 5.3 Badge and Access Control, Gates     | Test                      |
| FR-304     | F-03 Badge and Access Control     | 5.3 Badge and Access Control, Gates     | Test                      |
| FR-305     | F-03 Badge and Access Control     | 5.3 Badge and Access Control, Gates     | Test                      |
| FR-401     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-402     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-403     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-404     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-405     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-406     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-407     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-408     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-409     | F-04 Forum Programme              | 5.4 Forum Programme and Sessions        | Test                      |
| FR-501     | F-05 Bookings and Attendance      | 5.5 Bookings and Attendance             | Test                      |
| FR-502     | F-05 Bookings and Attendance      | 5.5 Bookings and Attendance             | Test                      |
| FR-503     | F-05 Bookings and Attendance      | 5.5 Bookings and Attendance             | Test                      |
| FR-504     | F-05 Bookings and Attendance      | 5.5 Bookings and Attendance             | Test                      |
| FR-505     | F-05 Bookings and Attendance      | 5.5 Bookings and Attendance             | Test                      |
| FR-506     | F-05 Bookings and Attendance      | 5.5 Bookings and Attendance             | Test                      |
| FR-601     | F-06 Exhibition                   | 5.6 Exhibition                          | Test                      |
| FR-602     | F-06 Exhibition                   | 5.6 Exhibition                          | Test                      |
| FR-603     | F-06 Exhibition                   | 5.6 Exhibition                          | Test                      |
| FR-604     | F-06 Exhibition                   | 5.6 Exhibition                          | Test                      |
| FR-605     | F-06 Exhibition                   | 5.6 Exhibition                          | Test                      |
| FR-701     | F-07 Engagement and Live Sessions | 5.7 Engagement, Live, Q and A, Ratings  | Test                      |
| FR-702     | F-07 Engagement and Live Sessions | 5.7 Engagement, Live, Q and A, Ratings  | Test                      |
| FR-703     | F-07 Engagement and Live Sessions | 5.7 Engagement, Live, Q and A, Ratings  | Test                      |
| FR-704     | F-07 Engagement and Live Sessions | 5.7 Engagement, Live, Q and A, Ratings  | Test                      |
| FR-705     | F-07 Engagement and Live Sessions | 5.7 Engagement, Live, Q and A, Ratings  | Test                      |
| FR-706     | F-07 Engagement and Live Sessions | Withdrawn from scope, no design content | Not applicable, withdrawn |
| FR-707     | F-07 Engagement and Live Sessions | Withdrawn from scope, no design content | Not applicable, withdrawn |
| FR-708     | F-07 Engagement and Live Sessions | 5.7 Engagement, Live, Q and A, Ratings  | Test                      |
| FR-709     | F-07 Engagement and Live Sessions | 5.7 Engagement, Live, Q and A, Ratings  | Test                      |
| FR-710     | F-07 Engagement and Live          | 5.7 Engagement, Live, Q and A, Ratings  | Test                      |

| Identifier | Service                              | Designed in                                                 | Verification method       |
|------------|--------------------------------------|-------------------------------------------------------------|---------------------------|
|            | Sessions                             |                                                             |                           |
| FR-801     | F-08 Networking and Cognitive AI     | 5.8 Networking and Cognitive AI                             | Test                      |
| FR-802     | F-08 Networking and Cognitive AI     | 5.8 Networking and Cognitive AI                             | Test                      |
| FR-803     | F-08 Networking and Cognitive AI     | 5.8 Networking and Cognitive AI                             | Test                      |
| FR-804     | F-08 Networking and Cognitive AI     | 5.8 Networking and Cognitive AI, and 5.13 Business Meetings | Test                      |
| FR-805     | F-08 Networking and Cognitive AI     | 5.8 Networking and Cognitive AI                             | Test                      |
| FR-806     | F-08 Networking and Cognitive AI     | 5.8 Networking and Cognitive AI                             | Test                      |
| FR-807     | F-08 Networking and Cognitive AI     | Withdrawn from scope, no design content                     | Not applicable, withdrawn |
| FR-808     | F-08 Networking and Cognitive AI     | 5.8 Networking and Cognitive AI                             | Test                      |
| FR-901     | F-09 Notifications                   | 5.9 Notifications                                           | Test                      |
| FR-902     | F-09 Notifications                   | 5.9 Notifications                                           | Test                      |
| FR-903     | F-09 Notifications                   | 5.9 Notifications                                           | Test                      |
| FR-904     | F-09 Notifications                   | 5.9 Notifications                                           | Test                      |
| FR-1001    | F-10 Media, News and Archive         | 5.10 Media, News and Archive                                | Test                      |
| FR-1002    | F-10 Media, News and Archive         | 5.10 Media, News and Archive                                | Test                      |
| FR-1003    | F-10 Media, News and Archive         | 5.10 Media, News and Archive                                | Test                      |
| FR-1004    | F-10 Media, News and Archive         | 5.10 Media, News and Archive                                | Test                      |
| FR-1005    | F-10 Media, News and Archive         | 5.10 Media, News and Archive                                | Test                      |
| FR-1006    | F-10 Media, News and Archive         | 5.10 Media, News and Archive                                | Test                      |
| FR-1101    | F-11 Statistics                      | 5.11 Statistics and Dashboards                              | Test                      |
| FR-1102    | F-11 Statistics                      | 5.11 Statistics and Dashboards                              | Test                      |
| FR-1103    | F-11 Statistics                      | 5.11 Statistics and Dashboards                              | Test                      |
| FR-1104    | F-11 Statistics                      | 5.11 Statistics and Dashboards                              | Test                      |
| FR-1201    | F-12 Control Panel and Configuration | 5.12 Control Panel Configuration                            | Test                      |
| FR-1202    | F-12 Control Panel and Configuration | 5.12 Control Panel Configuration                            | Test                      |
| FR-1203    | F-12 Control Panel and Configuration | 5.12 Control Panel Configuration                            | Test                      |
| FR-1204    | F-12 Control Panel and Configuration | 5.12 Control Panel Configuration                            | Test                      |
| FR-1205    | F-12 Control Panel and Configuration | 5.12 Control Panel Configuration                            | Test                      |
| FR-1206    | F-12 Control Panel and Configuration | 5.12 Control Panel Configuration                            | Test                      |
| FR-1207    | F-12 Control Panel and Configuration | 5.12 Control Panel Configuration                            | Test                      |
| FR-1208    | F-12 Control Panel and Configuration | 5.12 Control Panel Configuration                            | Test                      |



## 8.2 Business Rules, Non-functional and Operational Requirements

Each row names the section of this document that designs the requirement.

| Identifier | Kind           | Designed in                               | Verification method       |
|------------|----------------|-------------------------------------------|---------------------------|
| BR-01      | Business rule  | 2.1.5                                     | Test                      |
| BR-02      | Business rule  | 5.1                                       | Test                      |
| BR-03      | Business rule  | 5.2                                       | Test                      |
| BR-04      | Business rule  | 5.3                                       | Test                      |
| BR-05      | Business rule  | 5.5                                       | Test                      |
| BR-06      | Business rule  | 5.5 and 2.2.3                             | Test                      |
| BR-07      | Business rule  | 5.7                                       | Test                      |
| BR-08      | Business rule  | Withdrawn from scope, stated in section 5 | Not applicable, withdrawn |
| BR-09      | Business rule  | 5.13                                      | Test                      |
| BR-10      | Business rule  | 5.7                                       | Test                      |
| BR-11      | Business rule  | 5.12                                      | Test                      |
| BR-12      | Business rule  | 7.1.3                                     | Test                      |
| BR-13      | Business rule  | 6.3 and 7.1.3                             | Test                      |
| BR-14      | Business rule  | 5.2 and 5.12                              | Test                      |
| NFR-01     | Non-functional | 2.1.4 and 7.1.3                           | Test and analysis         |
| NFR-02     | Non-functional | 2.1.5                                     | Test and analysis         |
| NFR-03     | Non-functional | 7.1.3                                     | Test and analysis         |
| NFR-04     | Non-functional | 7.2.5                                     | Test and analysis         |
| NFR-05     | Non-functional | 7.1.3                                     | Test and analysis         |
| NFR-06     | Non-functional | 4.1, 4.2 and 4.3                          | Test and analysis         |
| NFR-07     | Non-functional | 4.3                                       | Test and analysis         |
| NFR-08     | Non-functional | 7.2.1                                     | Test and analysis         |
| NFR-09     | Non-functional | 7.1 and 7.2.5                             | Test and analysis         |
| NFR-10     | Non-functional | 5.12                                      | Test and analysis         |
| NFR-11     | Non-functional | 6.3 and 7.1.3                             | Test and analysis         |
| NFR-12     | Non-functional | 1.2.1 and 2.1.1                           | Test and analysis         |
| OP-01      | Operational    | 7.1                                       | Demonstration             |
| OP-02      | Operational    | 6.1 and 6.3                               | Demonstration             |
| OP-03      | Operational    | 7.2.5                                     | Demonstration             |
| OP-04      | Operational    | 7.1.3                                     | Demonstration             |
| OP-05      | Operational    | 2.1.3                                     | Demonstration             |
| OP-06      | Operational    | 7.2.5                                     | Demonstration             |
| OP-07      | Operational    | 5.2 and 5.12                              | Demonstration             |
| OP-08      | Operational    | 5.9                                       | Demonstration             |
| OP-09      | Operational    | 5.7 and 5.6                               | Demonstration             |
| OP-10      | Operational    | 2.1.3                                     | Demonstration             |
| OP-11      | Operational    | 2.1.2 and 2.1.4                           | Demonstration             |
| OP-12      | Operational    | 5.8                                       | Demonstration             |
| OP-13      | Operational    | 7.1.3                                     | Demonstration             |
| OP-14      | Operational    | 5.9                                       | Demonstration             |
| OP-15      | Operational    | 1.2.1 and 2.1.1                           | Demonstration             |
| OP-16      | Operational    | 5.7                                       | Demonstration             |